

# Protocolo de ultrassom para exame de urgência: Guia passo a passo

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 [pocus101.com/rush-exam-ultrasound-protocol-step-by-step-guide](https://pocus101.com/rush-exam-ultrasound-protocol-step-by-step-guide)

Vi Dinh

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O choque indiferenciado é uma apresentação clínica comum, mas pode ser difícil determinar a etiologia dessa emergência potencialmente fatal e que exige intervenção imediata. Os exames laboratoriais e de imagem convencionais são demorados e podem não ser úteis para distinguir entre os diferentes tipos de choque (cardiogênico, hipovolêmico, obstrutivo e distributivo).

A implementação do ultrassom à beira do leito, no entanto, permite uma avaliação rápida do estado hemodinâmico do paciente. Especificamente, o exame “RUSH” (Ultrassom Rápido para Choque e Hipotensão), introduzido por Weingart et al. em 2006, foi desenvolvido para fornecer uma avaliação rápida à beira do leito de um paciente com choque indiferenciado. Com o uso do protocolo RUSH, as possíveis etiologias do choque podem ser rapidamente identificadas.

Uma dúvida frequente é qual a diferença entre o exame RUSH e o [exame FAST/eFAST](#) . O exame FAST é utilizado em contextos de trauma para determinar a origem de uma possível hemorragia. Já o exame RUSH é um protocolo de ultrassom mais abrangente, utilizado em qualquer paciente com hipotensão não diferenciada.

Em termos simples, este protocolo aborda os componentes do sistema cardiopulmonar, resumidos no mnemônico do Exame RUSH: **HI MAP** (veja abaixo). Quando um paciente apresenta uma “ *PAM baixa* ”, nosso objetivo é corrigir o problema e levá-lo a uma “ *PAM alta* ”. Portanto, utilizaremos esse mnemônico como uma ferramenta útil para rápida memorização durante a avaliação dos pacientes.

Índice

## O mnemônico **HI MAP**

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A principal indicação do exame de ultrassom RUSH é avaliar rapidamente qualquer paciente com choque e hipotensão indiferenciados.

Segue abaixo uma tabela simples que resume as patologias que podem ser encontradas usando o mnemônico HI MAP do protocolo de ultrassom para exame RUSH.

|           | Órgão                    | Patologia                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>H</b>  | Coração                  | Fração de ejeção reduzida -> Insuficiência cardíaca sistólica<br>; Fração de ejeção hiperdinâmica -> Choque distributivo ou hipovolêmico<br>; Derrame pericárdico -> Tamponamento cardíaco;<br>Sobrecarga<br>do ventrículo direito (VD) -> Embolia pulmonar (EP);<br>Anormalidade regional da motilidade da parede -> Infarto do miocárdio; Débito<br>cardíaco baixo -> Choque cardiogênico, hipovolêmico ou obstrutivo;<br>Débito cardíaco alto -> Choque distributivo. |
| <b>EU</b> | Veia Cava Inferior (VCI) | VCI colapsável -> Choque hipovolêmico ou distributivo;<br>VCI não colapsável -> Choque obstrutivo ou cardiogênico.                                                                                                                                                                                                                                                                                                                                                       |
| <b>M</b>  | Exame Morison/eFAST      | Hemoperitônio/Hemotórax -> Choque Hemorrágico                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>UM</b> | Aorta                    | Aneurisma da aorta abdominal<br>Dissecção da aorta                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>P</b>  | Pulmonar                 | Pneumotórax                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

É importante lembrar que o exame RUSH não é uma avaliação abrangente de todos os componentes listados, mas sim uma avaliação rápida das principais patologias que podem causar hipotensão (e choque) em um paciente. Com a prática, esse exame pode ser concluído em apenas 2 a 5 minutos.

As limitações enfrentadas por este protocolo são as mesmas de qualquer exame de ultrassom: biotipo do paciente, posicionamento e limitações físicas que dificultam a obtenção de imagens ideais.

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## Preparação para exame de ultrassom URSH

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O exame de ultrassom RUSH geralmente é realizado com o paciente em decúbito dorsal.

**Transdutor** : A sonda phased array será o principal equipamento do protocolo RUSH (alguns autores têm defendido o uso da sonda curvilínea, dependendo da capacidade do equipamento). Caso seja indicado um exame de TVP (Trombose Venosa Profunda), será necessário utilizar a sonda linear.

**Pré-configuração** : o modo de exame eFAST ou o modo de exame abdominal serão suficientes.

**Posicionamento** : Sempre que possível, posicione o aparelho de ultrassom à direita do paciente. Isso permite que sua mão direita manobre a sonda e sua mão esquerda ajuste as configurações.

# Protocolo passo a passo para o exame RUSH

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## Passo 1: Coração

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Primeiramente, analisamos o coração. Em um contexto de hipotensão, precisamos saber o quão bem o coração está se contraindo e ejetando o sangue. O problema está no coração ou ele está tentando compensar algum problema em outro lugar? As principais patologias a serem avaliadas são insuficiência cardíaca sistólica, derrame pericárdico e tamponamento cardíaco, além de sobrecarga do ventrículo direito para embolia pulmonar.

Existem quatro vistas cardíacas principais, incluindo a vista paraesternal de eixo longo, a vista paraesternal de eixo curto, a vista apical de quatro câmaras e a vista subxifoide. É importante aprender todas essas vistas, visto que o biotipo dos pacientes pode variar bastante.

*Observação sobre ultrassonografia à beira do leito (POCUS 101):* Lembre-se, ao usar a predefinição eFAST ou abdominal, o marcador de orientação ficará no lado esquerdo da tela, em vez do lado direito, como na predefinição cardíaca. Isso exigirá que você gire o indicador da sonda 180 graus para obter as visualizações cardíacas adequadas.

[Clique aqui para um tutorial passo a passo sobre como realizar todas as imagens de ultrassom cardíaco!](#)

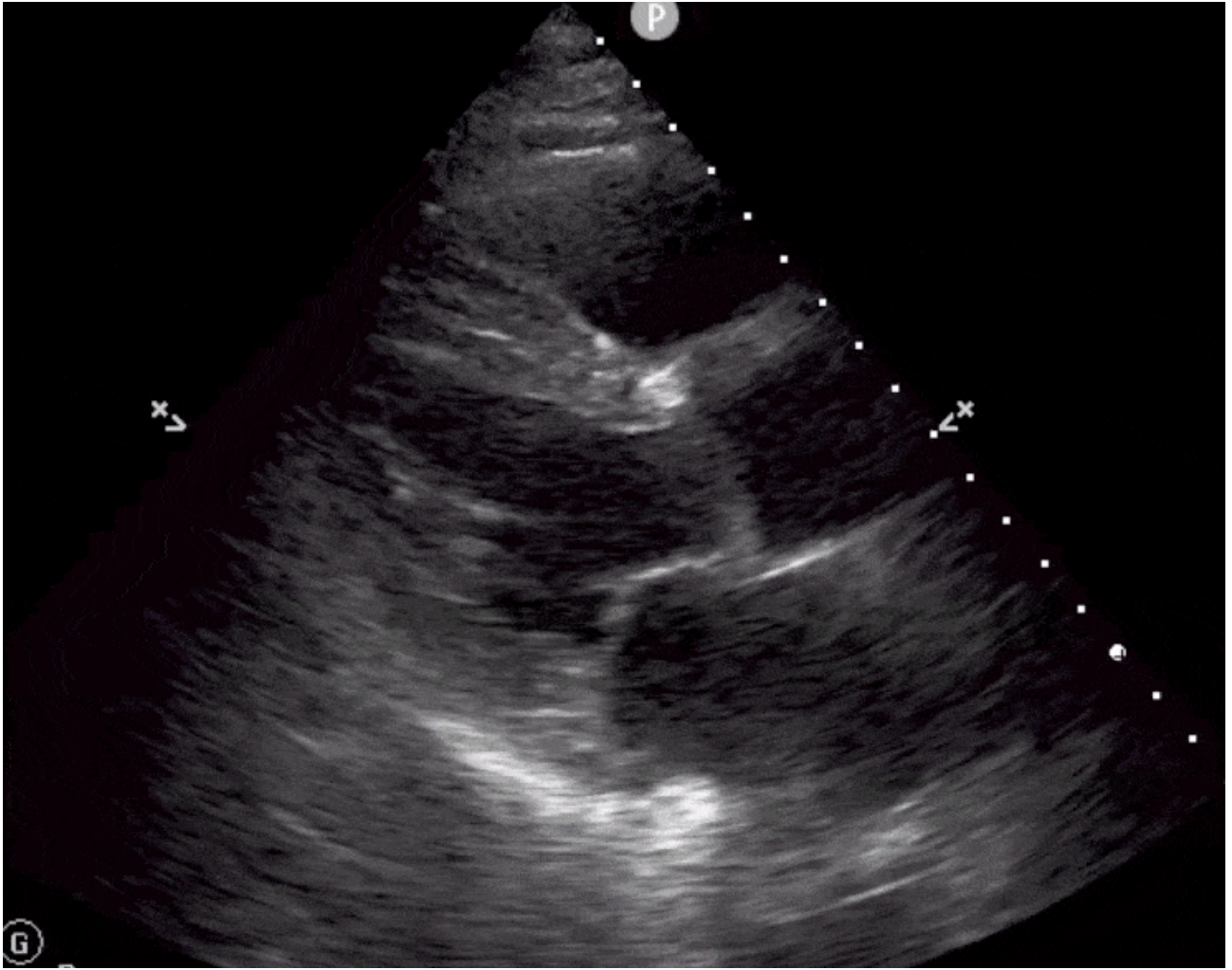
## Avaliação da fração de ejeção

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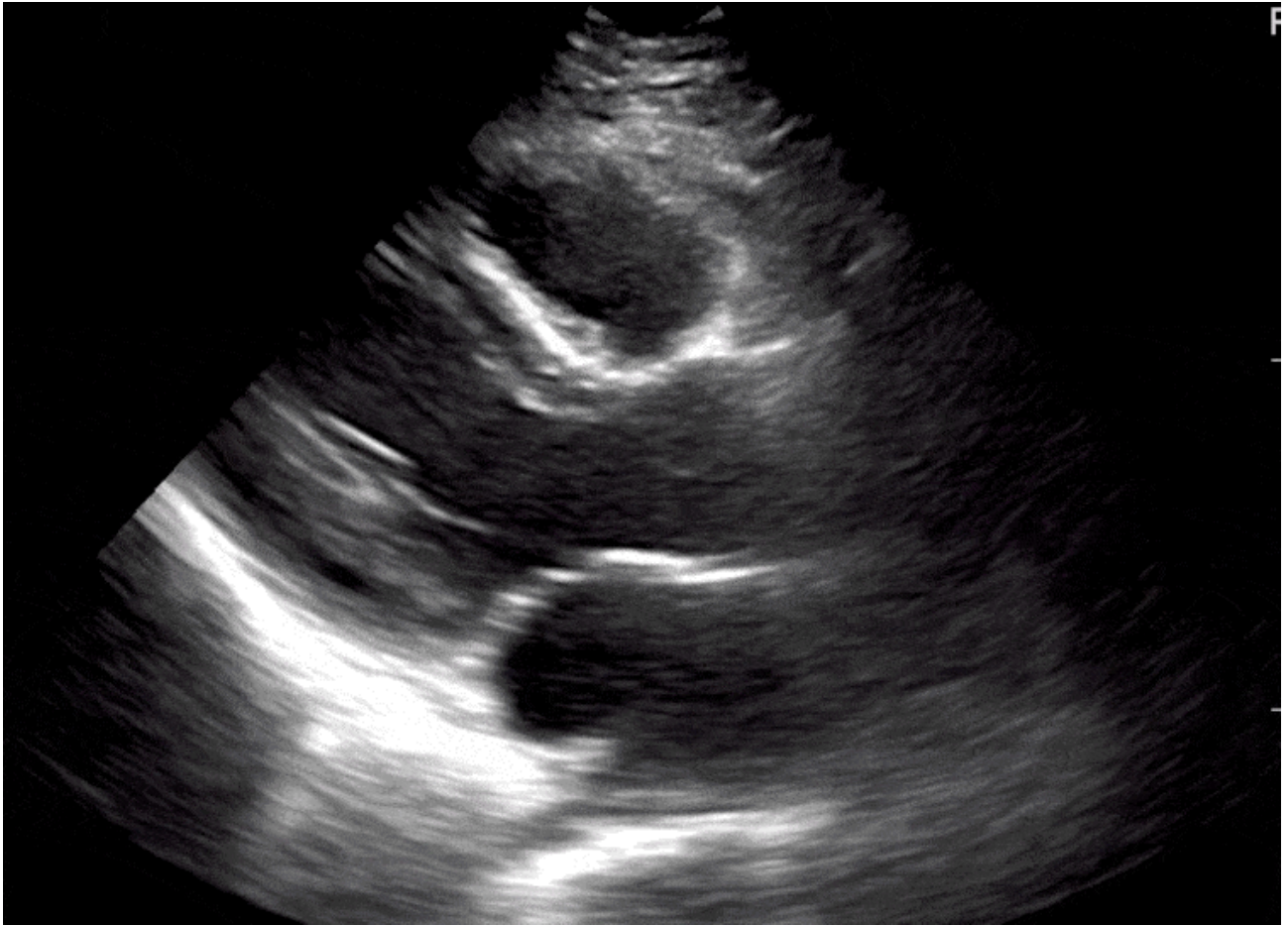
- Observe o ventrículo esquerdo ao longo do ciclo cardíaco. Ele é hiperdinâmico ou hipodinâmico? A contração é uniforme?
- Em seguida, observe a cúspide anterior da válvula mitral. Ela se move livremente e se aproxima do septo interventricular a cada enchimento diastólico? Caso contrário, a função contrátil do coração pode estar comprometida e o paciente pode estar apresentando uma exacerbação da insuficiência cardíaca sistólica, resultando em hipotensão.
- Se o coração apresentar hiperdinâmica, a causa da hipotensão pode estar relacionada à hipovolemia ou sepse. Isso pode ser determinado continuando a avaliação do paciente com o restante do exame RUSH.

[Clique aqui para um tutorial passo a passo sobre como avaliar a fração de ejeção do ventrículo esquerdo!](#)

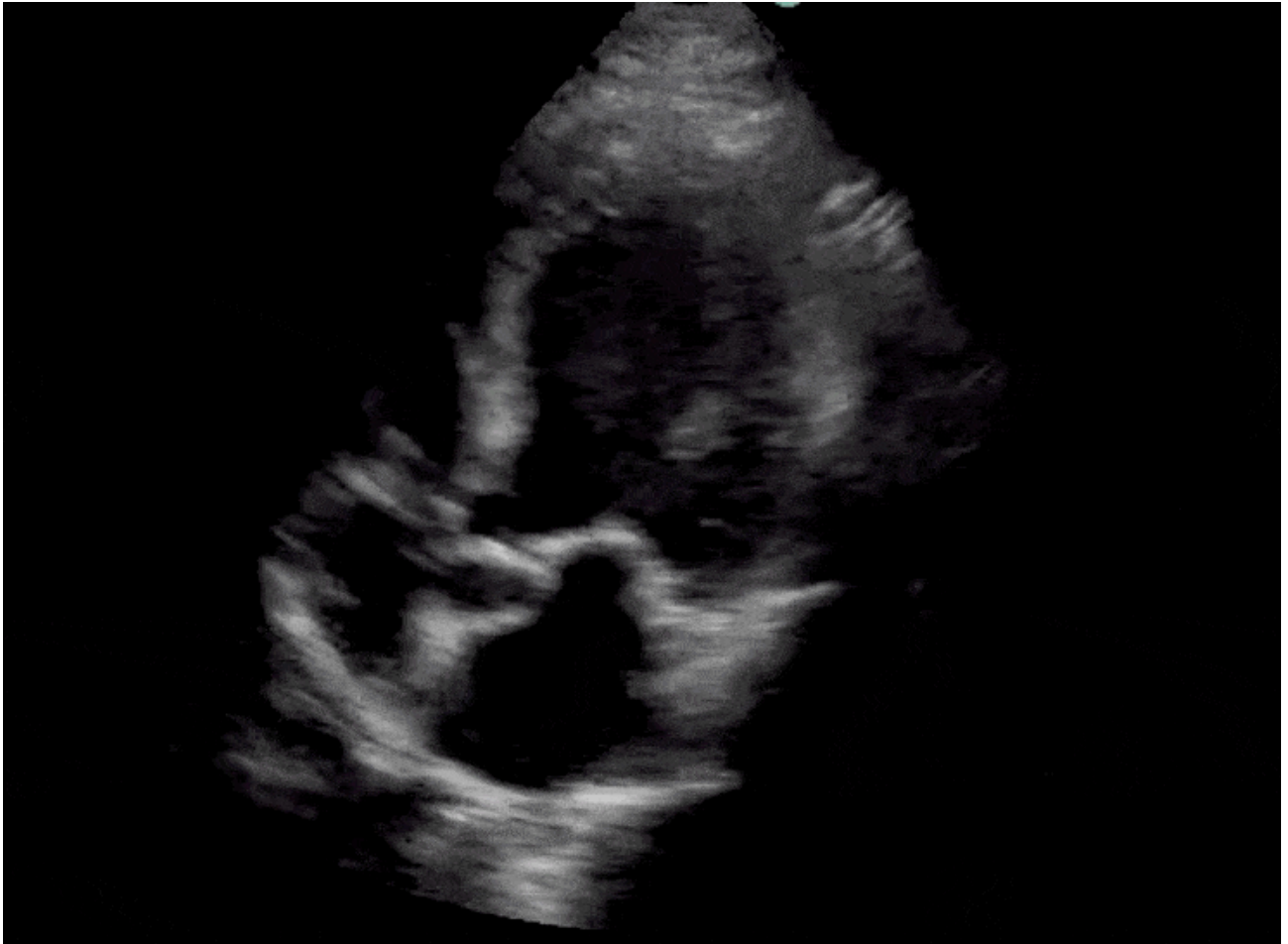
Aqui estão imagens de ultrassom cardíaco (ecocardiograma) de pacientes com diferentes graus de fração de ejeção, desde hiperdinâmica até gravemente reduzida:



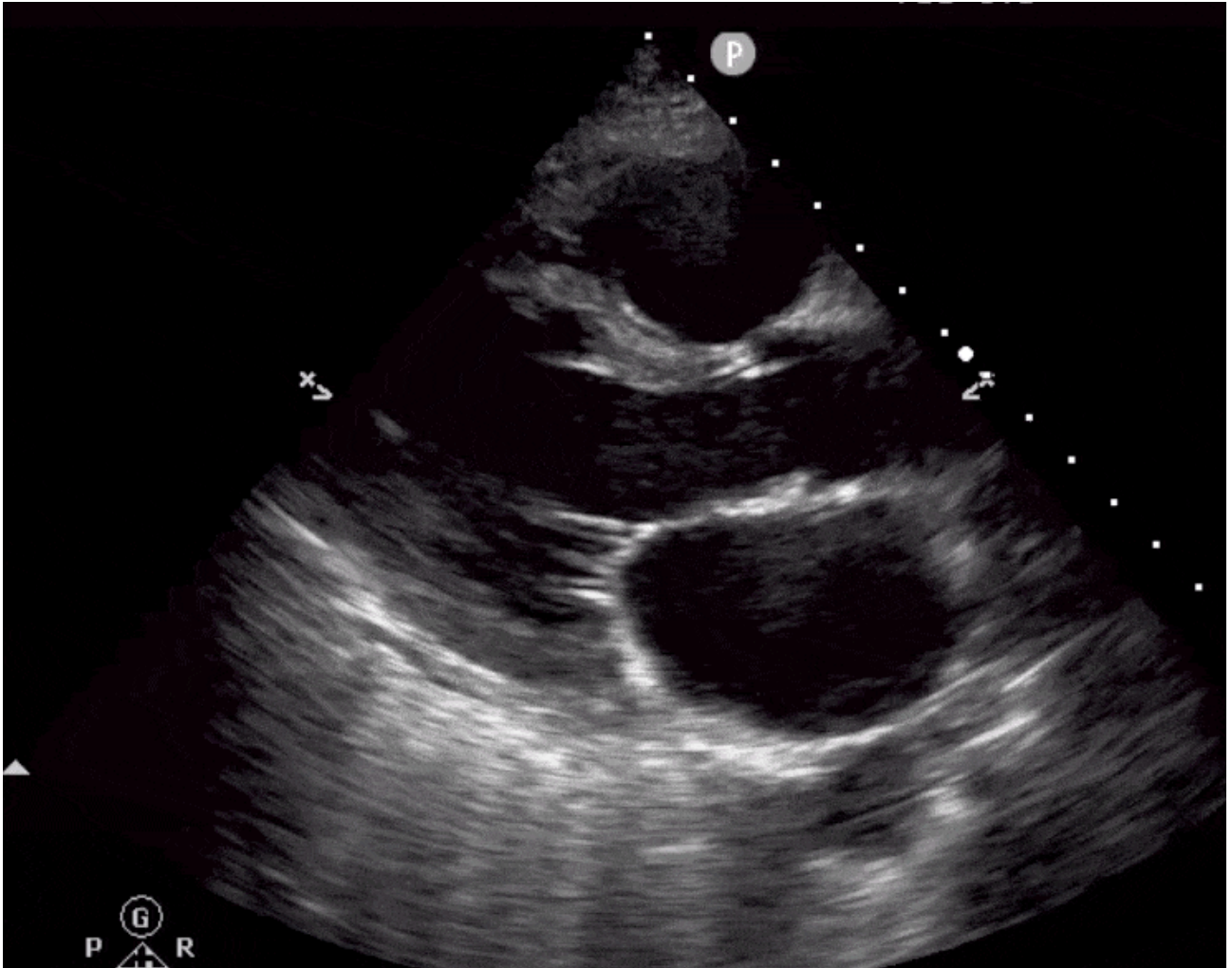
Fração de ejeção hiperdinâmica ( > 70%)



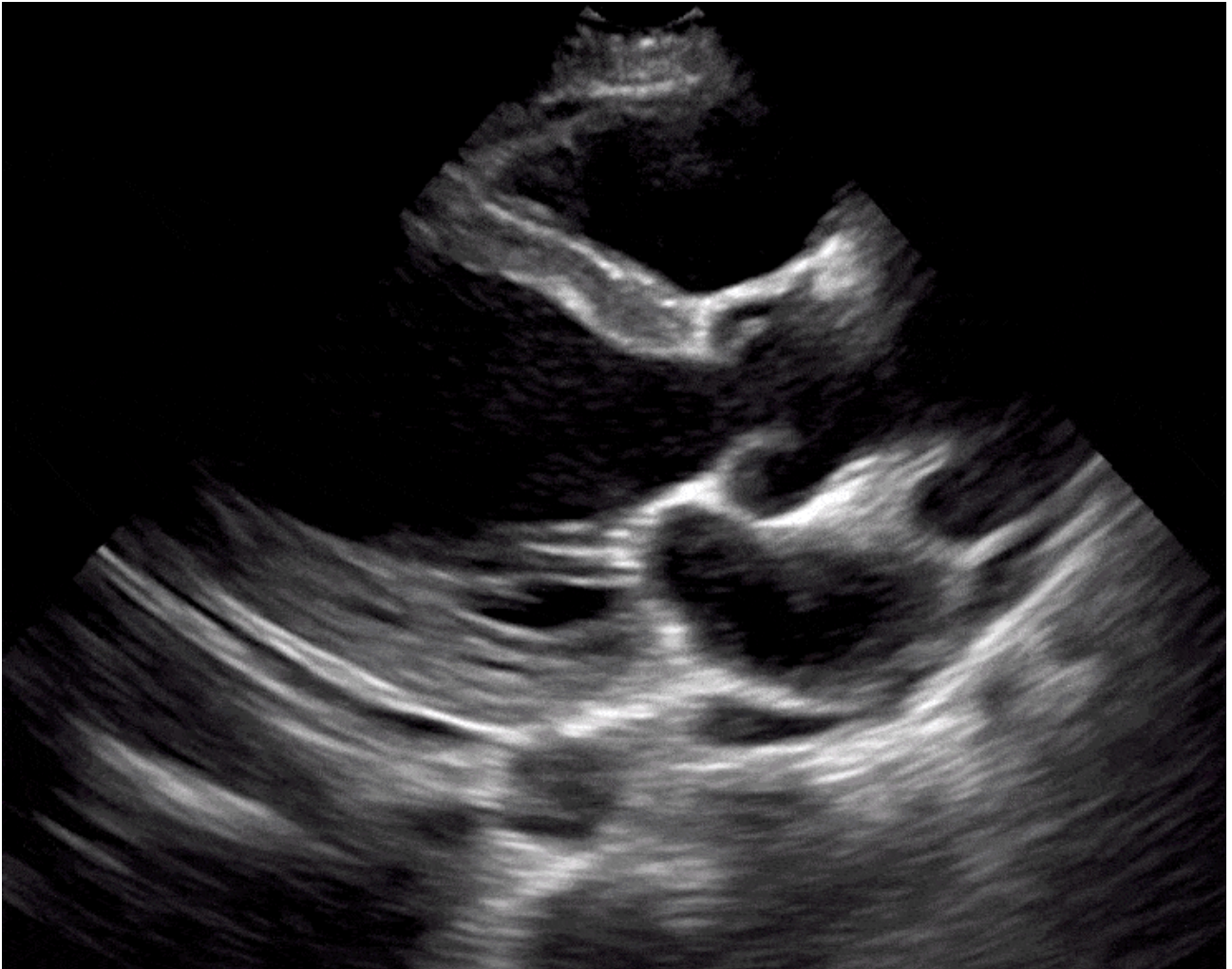
Fração de ejeção normal (55-69%)



Fração de ejeção levemente reduzida (45-54%)



Fração de ejeção moderadamente reduzida (30-44%)

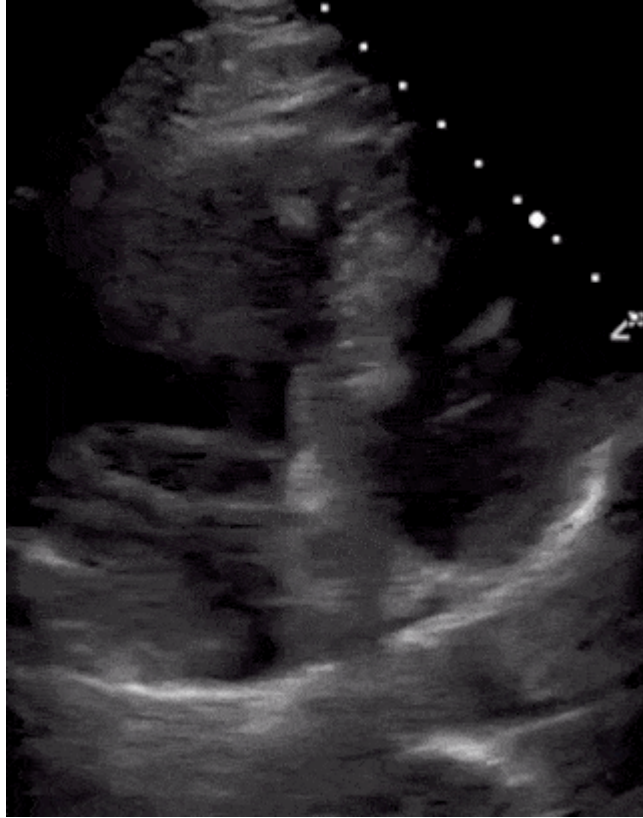


Fração de ejeção gravemente reduzida (<30%)

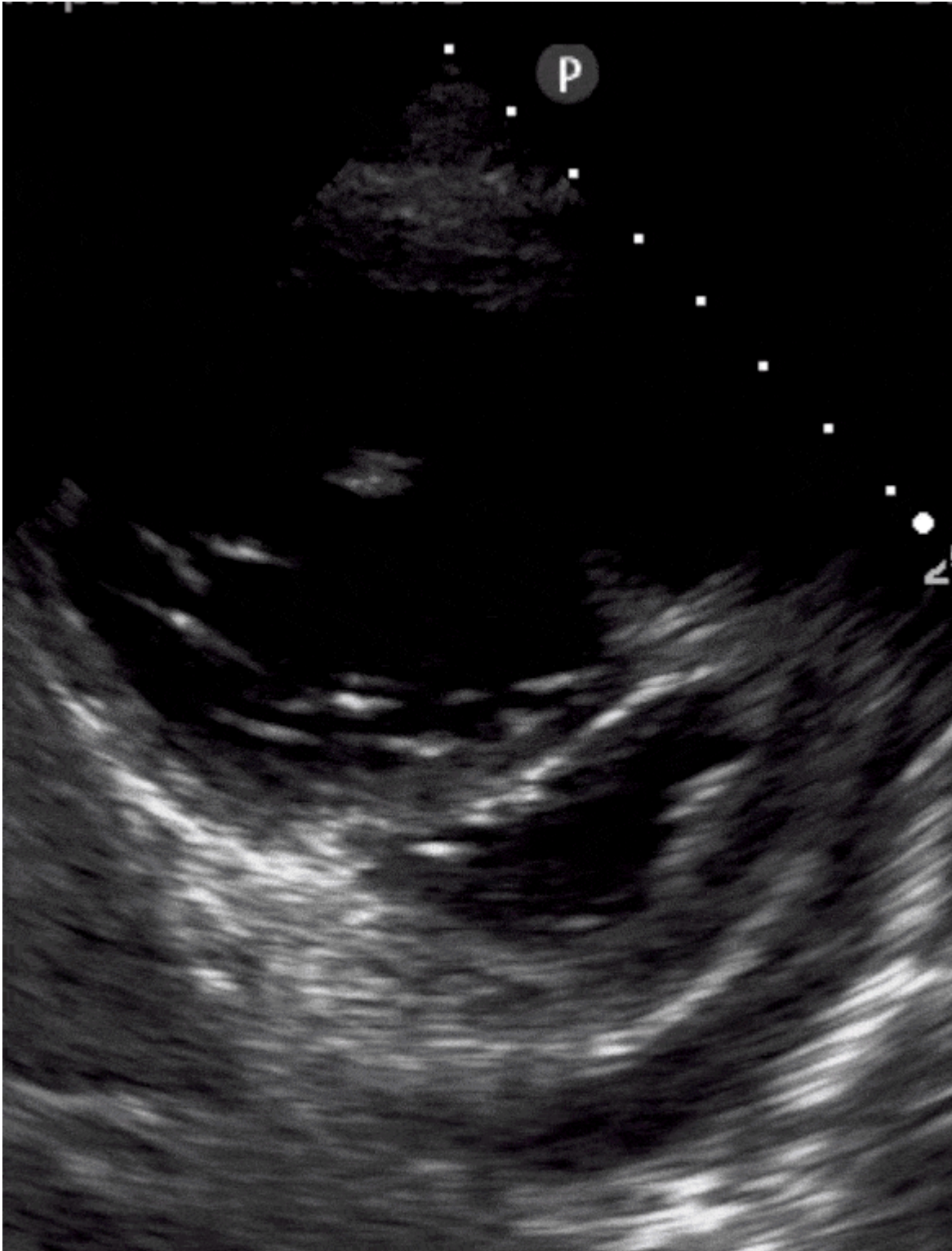
### Sobrecarga do ventrículo direito – embolia pulmonar

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- The RV chamber normally size should be about  $\frac{2}{3}$  the size of the LV chamber. If you notice an increase in this ratio, or bowing of the interventricular septum towards the left side of the heart, this may indicate RV strain (i.e. the right heart is having trouble ejecting, so as pressure builds the RV chamber enlarges and encroaches on the LV).
- Look out for **McConnell's Sign** which is hypokinesis of the right ventricular free wall with sparing of the Apex.
- You can also look for the “**D Sign**” on the parasternal short axis view and occurs when the increased pressure of the RV pushes the interventricular septum into the LV.  
[Click HERE to learn more about the “D Sign.”](#)



RV Strain with McConnell's Sign



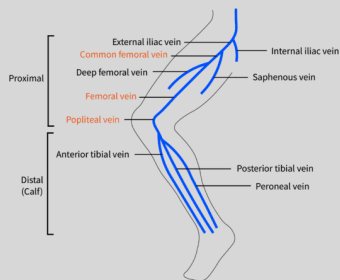
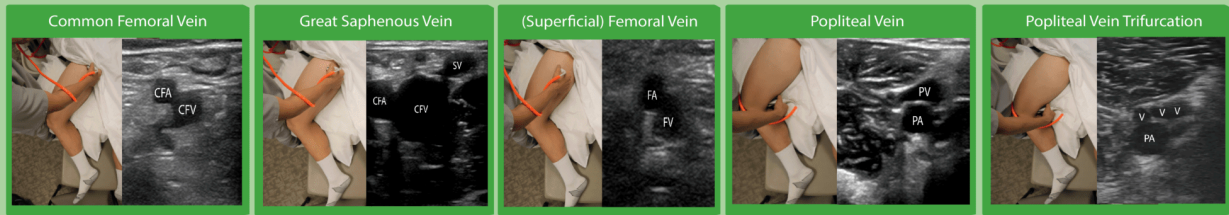
"D Sign"

- Note that the presence of any of the above-mentioned findings is suggestive of RV strain and **not specific to PE**.
- If any evidence of right heart strain is found, the exam should **proceed directly** to the evaluation of the legs to scan for a possible deep vein thrombosis by looking for a noncompressible vein. Findings of DVT in the setting of RV strain will greatly increase the chances the patient has a significant pulmonary embolism. [Click HERE to learn about how to diagnose DVT using POCUS.](#)

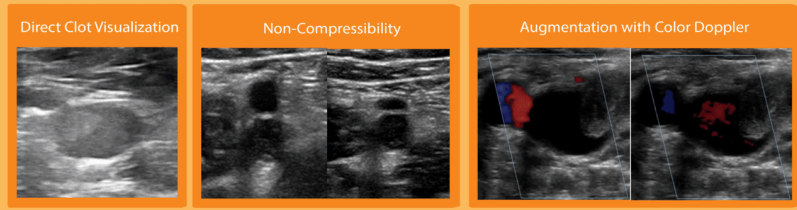
# Deep Vein Thrombosis Ultrasound Pocket Card

POCUS 101

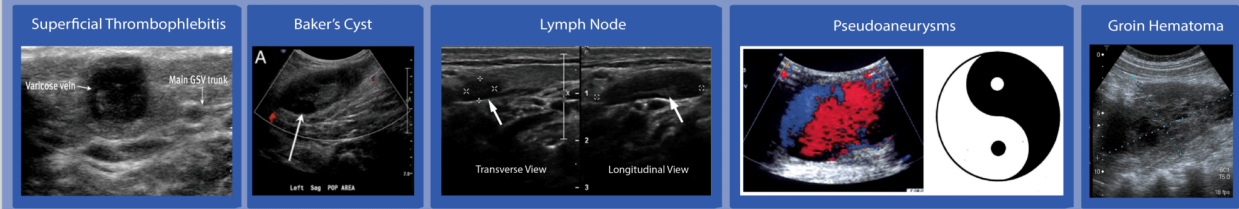
## Deep Vein Thrombosis Scanning Sites



## Deep Vein Thrombosis Pathology



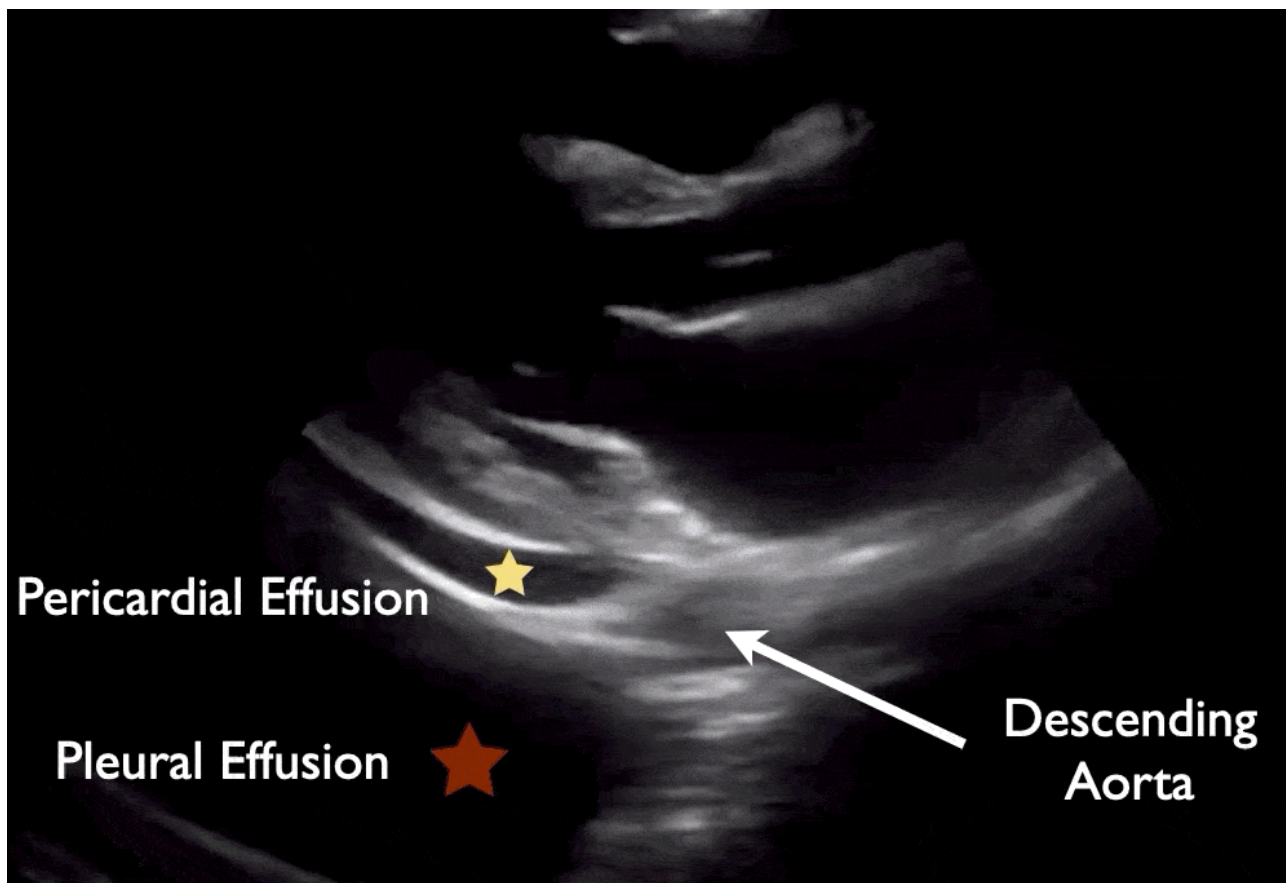
## Deep Vein Thrombosis False Positives



Download the DVT Ultrasound Pocket Card!

## Pericardial Effusion and Tamponade

- Pericardial effusion with Tamponade is another cardiac pathology you should screen for in the RUSH Protocol.
- Look at the border of the heart. Is there a significant anechoic border surrounding it? If so, a pericardial effusion is likely.
- Be sure to distinguish a pericardial effusion from a pleural effusion by identifying the location of the fluid with help from the descending aorta. Anechoic fluid anterior to the descending aorta is a pericardial effusion, whereas fluid posterior to the descending aorta is a pleural effusion.



Pericardial Effusion – PSLA View



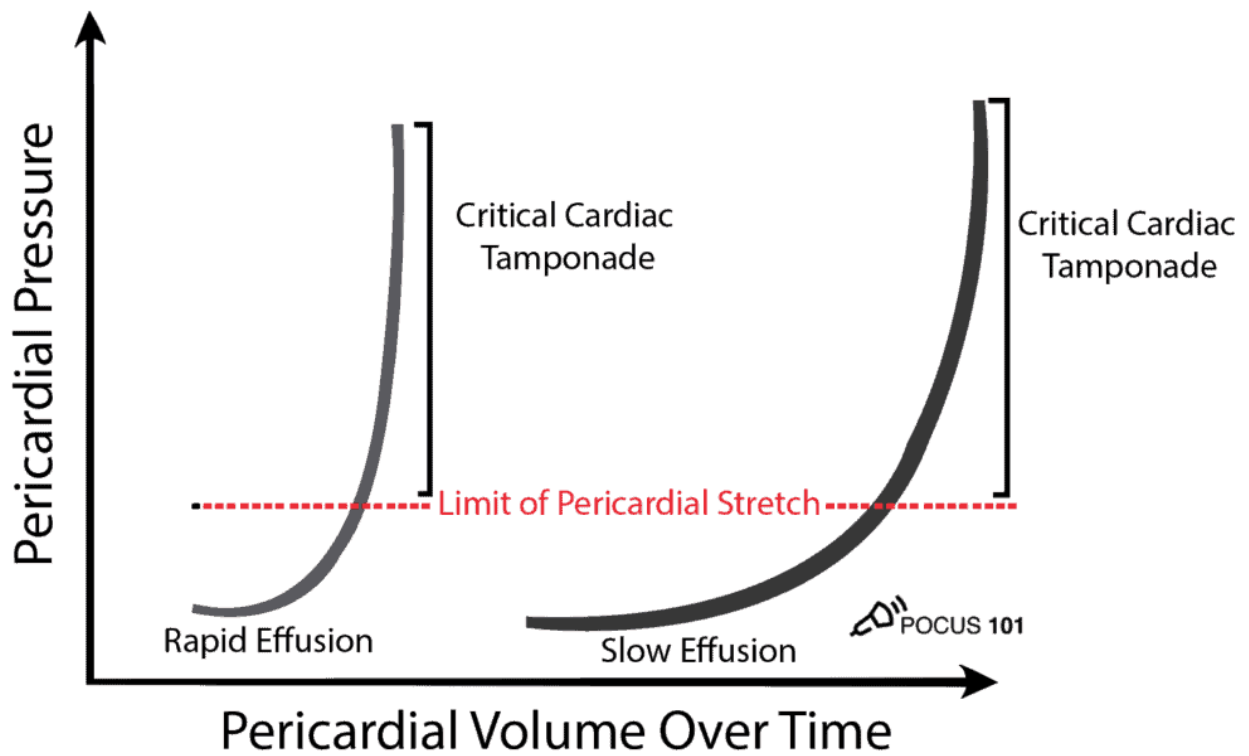
Pericardial Effusion -Subxiphoid View

Note that a pericardial fat pad can mimic a pleural effusion so it is important to distinguish between the two. Fat is usually more anterior, less mobile, and more echogenic than the fluid of an effusion.



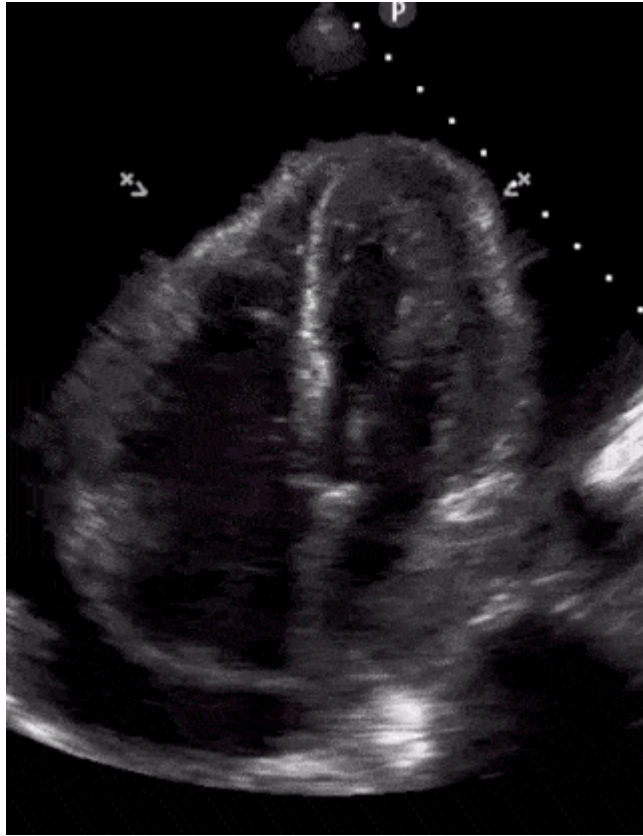
Pericardial Fat Pad

A pericardial effusion can lead to cardiac tamponade when the pressure from the effusion becomes high enough to impair cardiac filling of the right ventricle. It is important to remember that it is not the absolute size of the pericardial effusion but the rate of fluid accumulation that poses the greatest risk for cardiac tamponade.

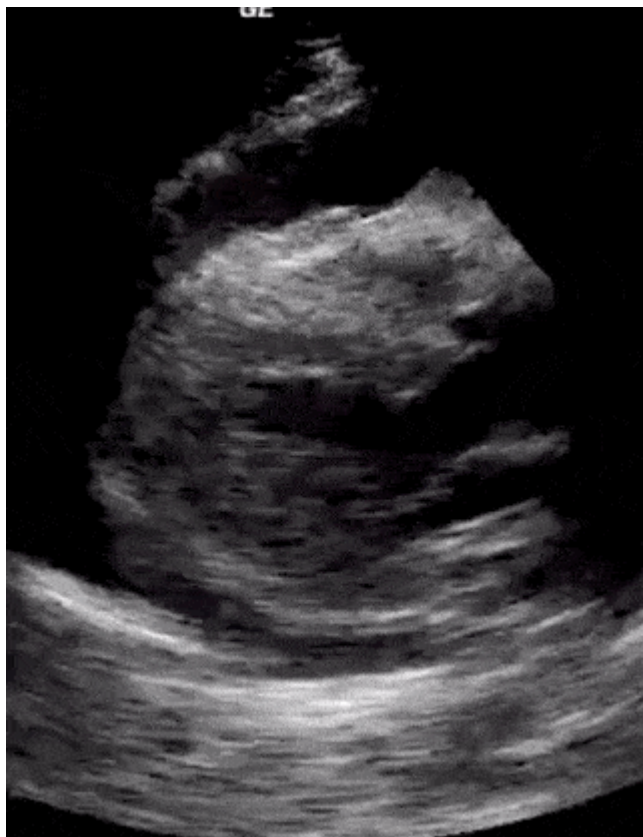


To distinguish between an effusion and tamponade on the RUSH Exam look for two signs: right atrial systolic collapse and right ventricular diastolic collapse.

- These are difficult to capture in real time, so freeze your image and scroll through the frame at a slower rate to determine when systole and diastole are occurring.
- Right atrial collapse is the earliest and the most sensitive finding.
- Right ventricular diastolic collapse is a later and more specific finding.



Pericardial Effusion with Tamponade – RA Systolic Collapse



Cardiac Tamponade – RV Diastolic Collapse

## Cardiac Output Assessment

You can also measure the patient's cardiac output during the RUSH Exam to help determine the type of Shock a patient is in.

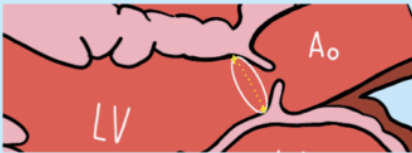
This is a more advanced technique and can be done by measuring the Left Ventricular Outflow Tract Diameter and the Velocity Time Integral (VTI) to calculate the cardiac output.

[Click HERE to for a Step By Step Tutorial on Measuring Cardiac Output.](#)

### MEASURING CARDIAC OUTPUT

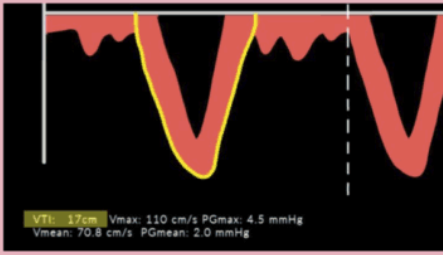
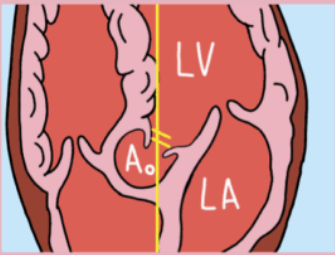
#### CALCULATE LVOT AREA

1. PARASTERNAL LONG AXIS VIEW
2. ZOOM INTO LVOT
3. MEASURE LVOT DIAMETER IN CM
4. CALCULATE LVOT AREA USING AREA OF A CIRCLE FORMULA


$$\text{LVOT AREA} = \pi \left( \frac{\text{cm}}{2} \right)^2$$

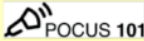
#### CALCULATE LVOT VTI

1. APICAL 5 CHAMBER VIEW
2. PLACE PULSE WAVE DOPPLER GATE AT LVOT
3. ACTIVATE PW DOPPLER
4. TRACE AROUND EJECTION WAVE
5. RECORD VTI IN CM



VTI: 17cm Vmax: 110 cm/s PGmax: 4.5 mmHg  
Vmean: 70.8 cm/s PGmean: 2.0 mmHg

#### CALCULATE CARDIAC OUTPUT

$$\text{SV} = \text{LVOT AREA} \times \text{LVOT VTI}$$
$$\text{CO} = (\text{LVOT AREA} \times \text{LVOT VTI}) \times \text{HR}$$
$$\text{CO (mL/min)} = \text{SV (mL/cycle)} \times \text{HR (bpm)}$$


[Download the Cardiac Output Pocket Guide HERE!](#)

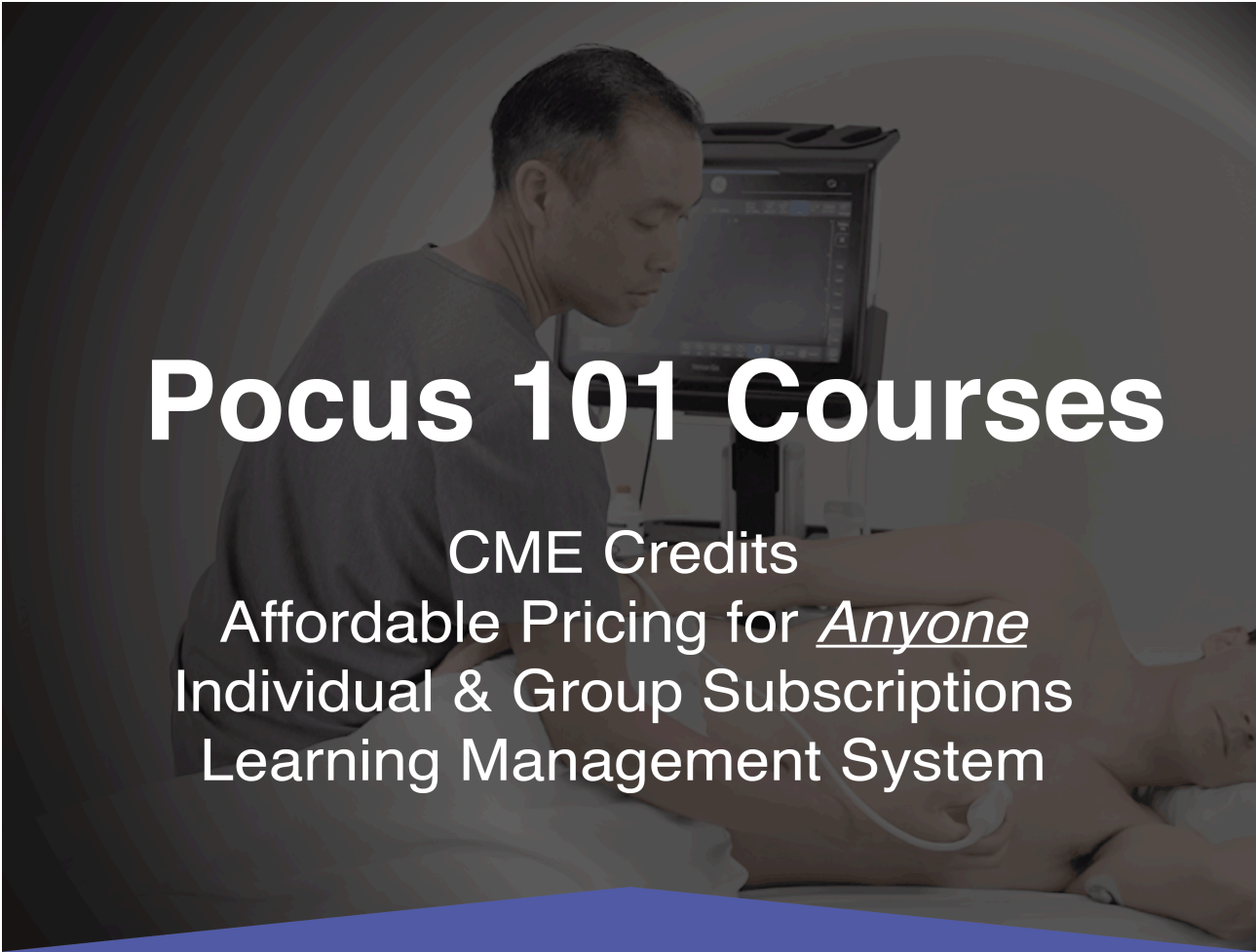
Measuring the cardiac output helps you differentiate the different types of shock, especially when you combine it with ejection fraction and IVC Measurements as you can see below:

| Type of Shock | LV Ejection Fraction | Cardiac Output | IVC            |
|---------------|----------------------|----------------|----------------|
| Distributive  | High                 | High           | Collapsible    |
| Obstructive   | Normal/High          | Low            | Noncollapsible |
| Cardiogenic   | Low                  | Low            | Noncollapsible |
| Hypovolemic   | High                 | Low            | Collapsible    |

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## Step 2: IVC

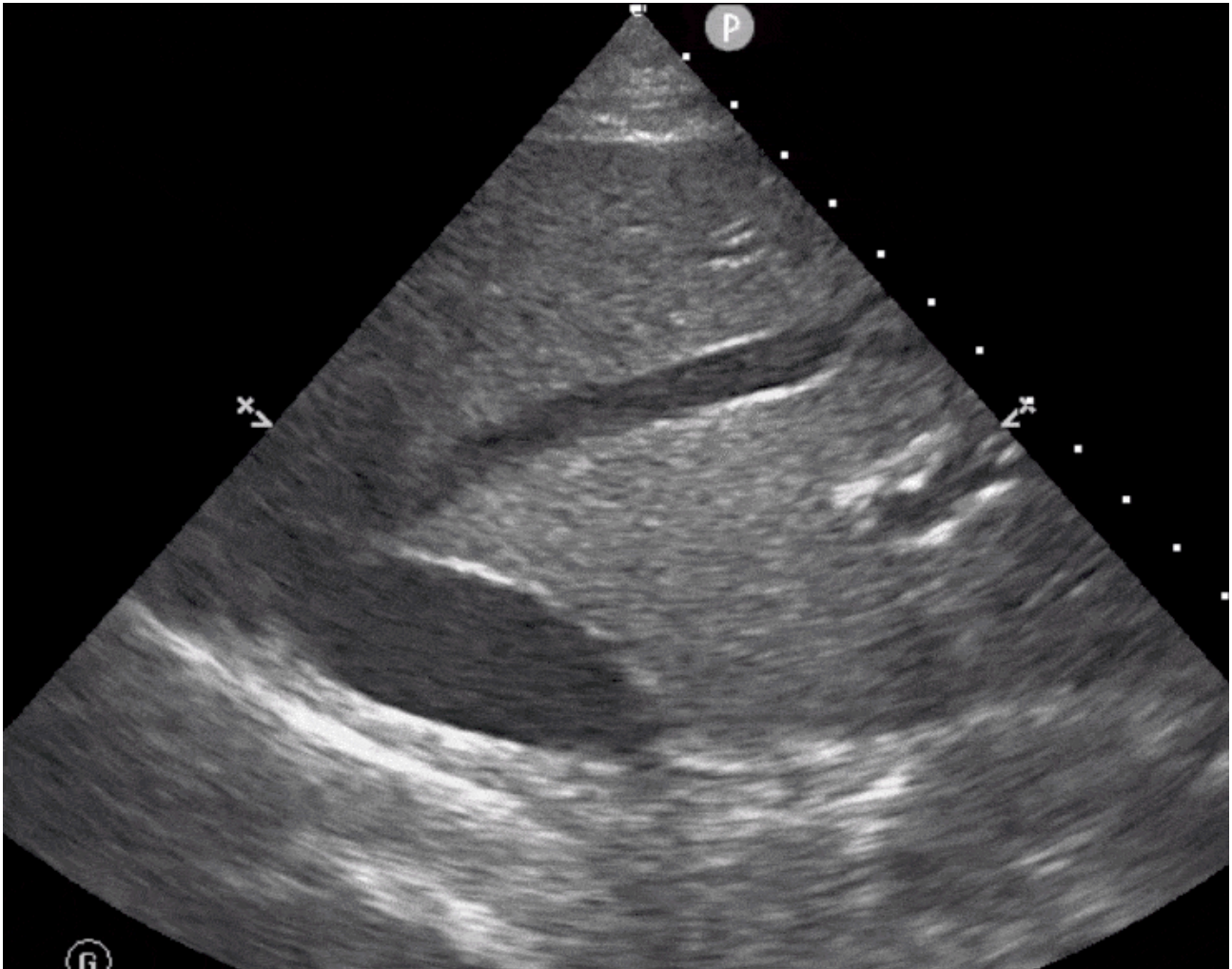
The next part of the RUSH Exam, we look at the Inferior Vena Cava (IVC) to assess the patient's central venous pressure (or right atrial pressure). Is there a leak somewhere causing hypotension? Is there fluid overload that is causing the heart to not pump adequately? In the context of shock, we also want to know whether CVP is high or low. This will further categorize the likely type of shock.

[Click here for a step-by-step tutorial on how to acquire the IVC view.](#)

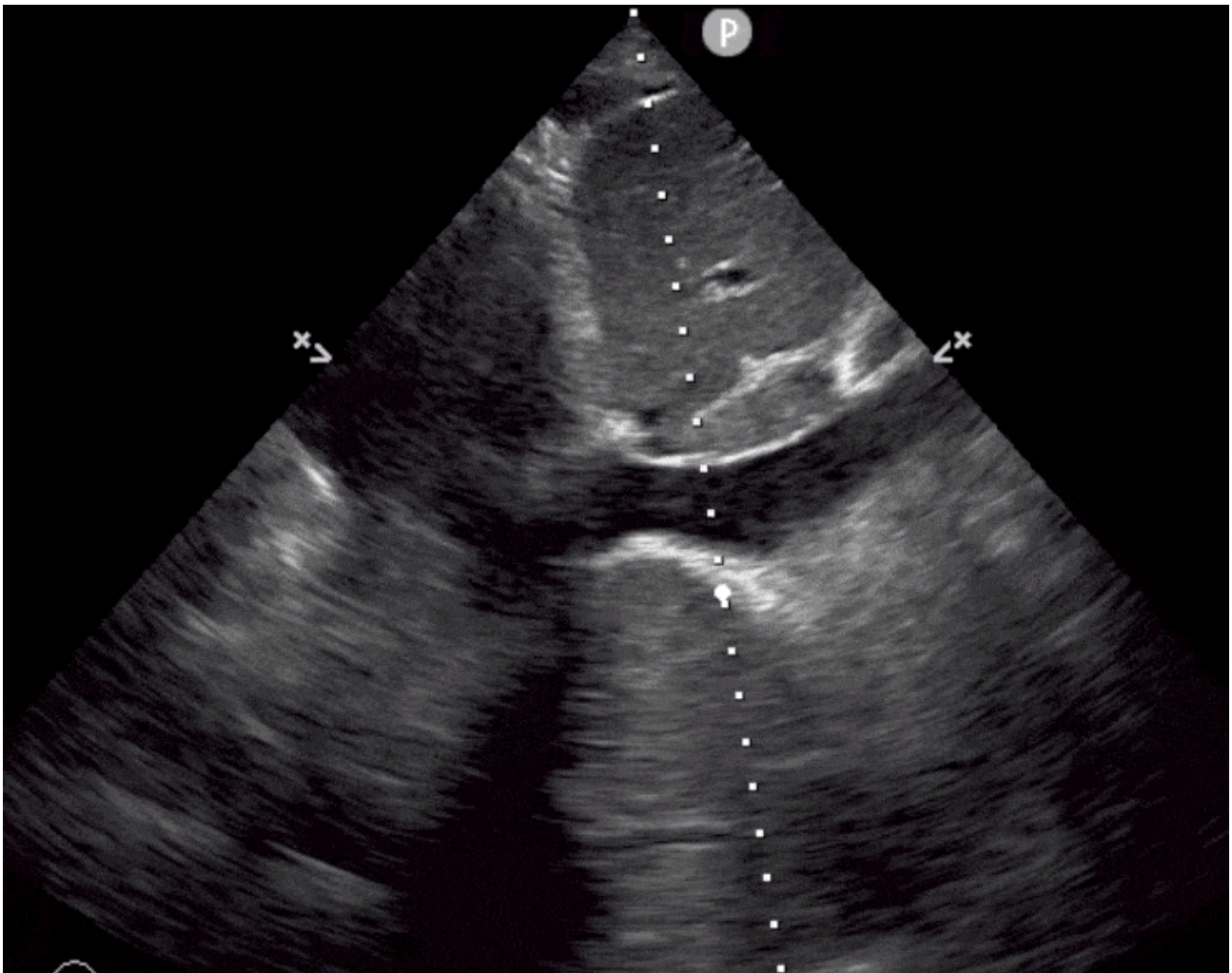
The IVC measurement is mostly used to assess fluid tolerance rather than fluid responsiveness. Using the IVC collapsibility Index (below), the diameter and collapsibility during inspiration or with a sniff test can be used to estimate CVP. Limitations include body habitus, increased intraabdominal pressure, etc.

| IVC Diameter (cm) | Percent Collapsible (%) | Estimated CVP (mmHg) |
|-------------------|-------------------------|----------------------|
| <1.5              | >50                     | 0-5                  |
| 1.5-2.5           | >50                     | 6-10                 |
| 1.5-2.5           | <50                     | 11-15                |
| >2.5              | <50                     | 16-20                |

Adapted from Kircher et al.



Dilated and Noncollapsible IVC – Suggesting HIGH CVP



Small and Collapsible IVC – Suggesting LOW CVP

- A high CVP suggested by a dilated and noncollapsible IVC may hint towards an obstructive or cardiogenic etiology.
- A low CVP suggested by a small and collapsible IVC may hint towards a distributive or hypovolemic etiology.

Once again here is an easy way to integrate LV ejection fraction, Cardiac Output, and IVC measurements:

| Type of Shock | LV Ejection Fraction | Cardiac Output | IVC            |
|---------------|----------------------|----------------|----------------|
| Distributive  | High                 | High           | Collapsible    |
| Obstructive   | Normal/High          | Low            | Noncollapsible |
| Cardiogenic   | Low                  | Low            | Noncollapsible |
| Hypovolemic   | High                 | Low            | Collapsible    |

### Step 3: Morison's Pouch/eFAST Exam

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The “M” in HIMAP of the RUSH Exam stand's for Morison's Pouch but essentially you will be performing an eFAST scan of the RUQ, LUQ, and pelvic views. We are looking for a leak into the abdomen (hemoperitoneum) or thorax (hemothorax) as the source of the patient's hypotension.

For a complete tutorial on the [eFAST Exam Click HERE](#).

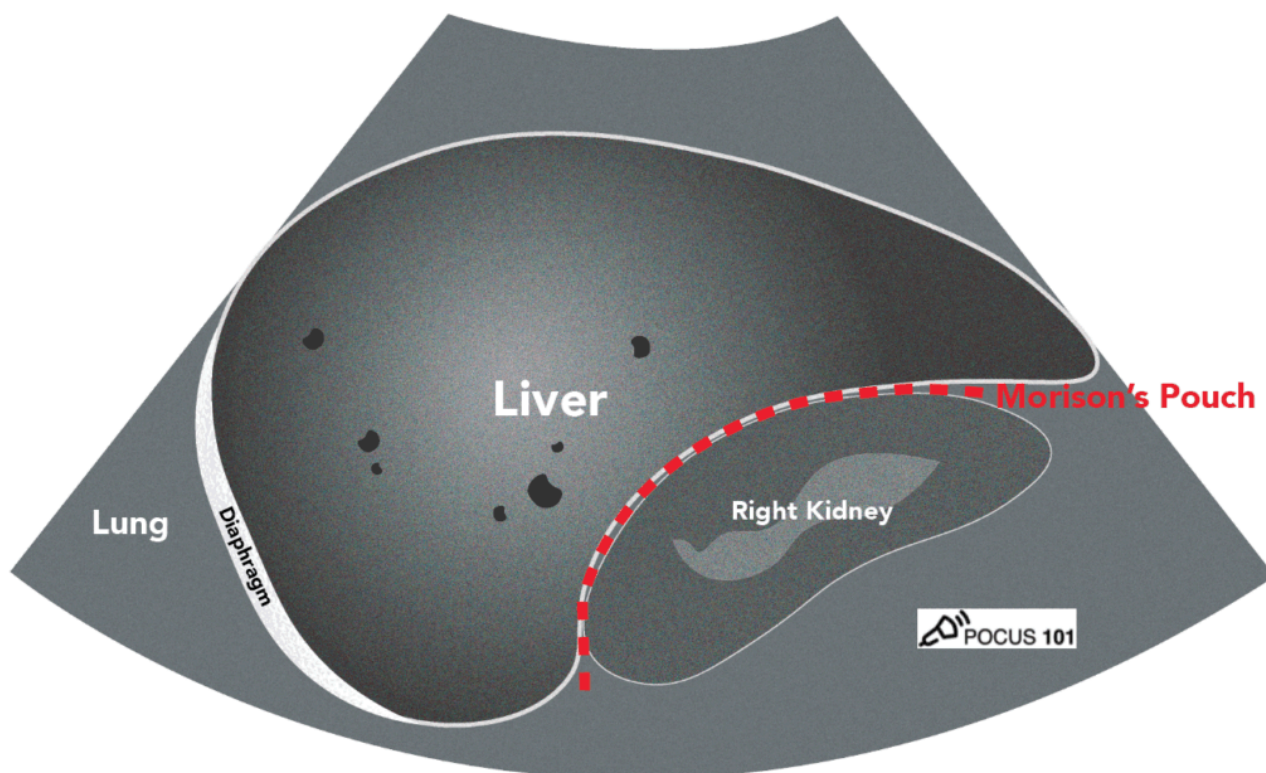
#### Hemoperitoneum

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##### Right Upper Quadrant (RUQ) – Hemoperitoneum

The three common locations for free fluid to accumulate in the RUQ of the eFAST scan are the:

- Hepatorenal Space or “Morison's Pouch”
- Caudal Tip of the Liver
- Suprahepatic Space



eFAST RUQ Morison's Pouch



Free Fluid at the Caudal Tip of the Liver



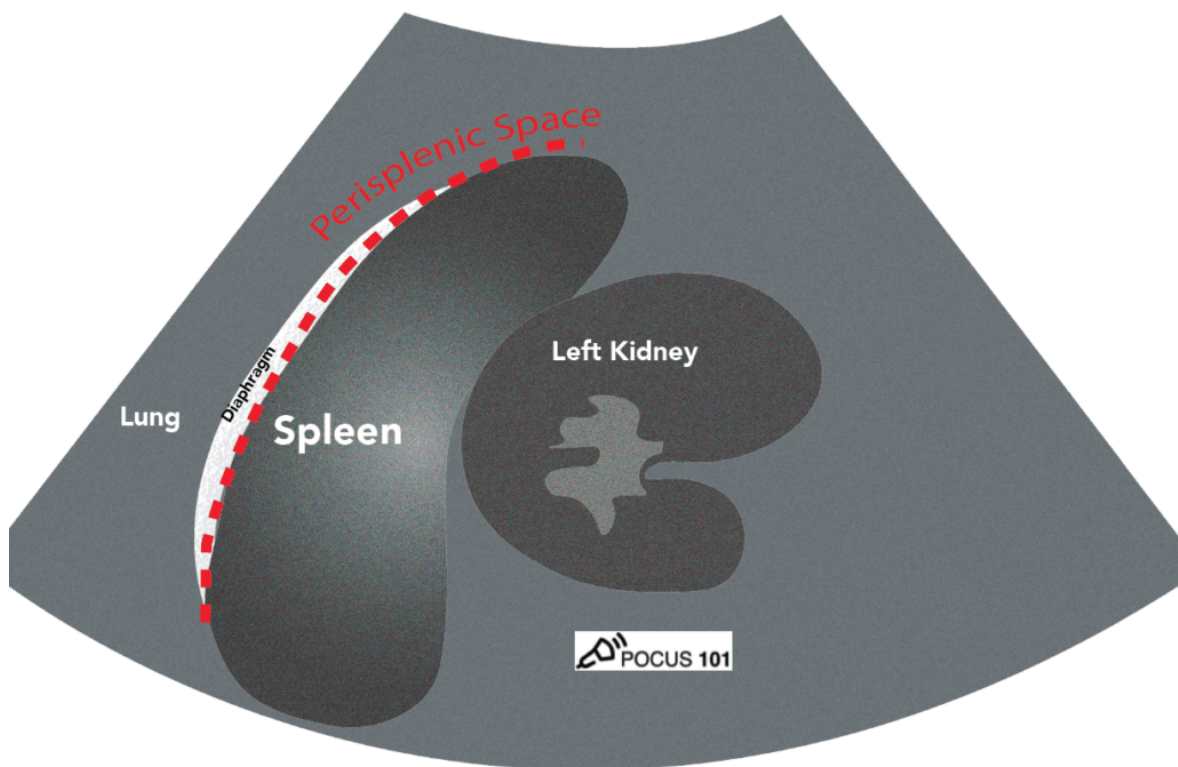
Free Fluid in Morrison's Pouch and Suprahepatic Space

### **Left Upper Quadrant (LUQ) – Hemoperitoneum**

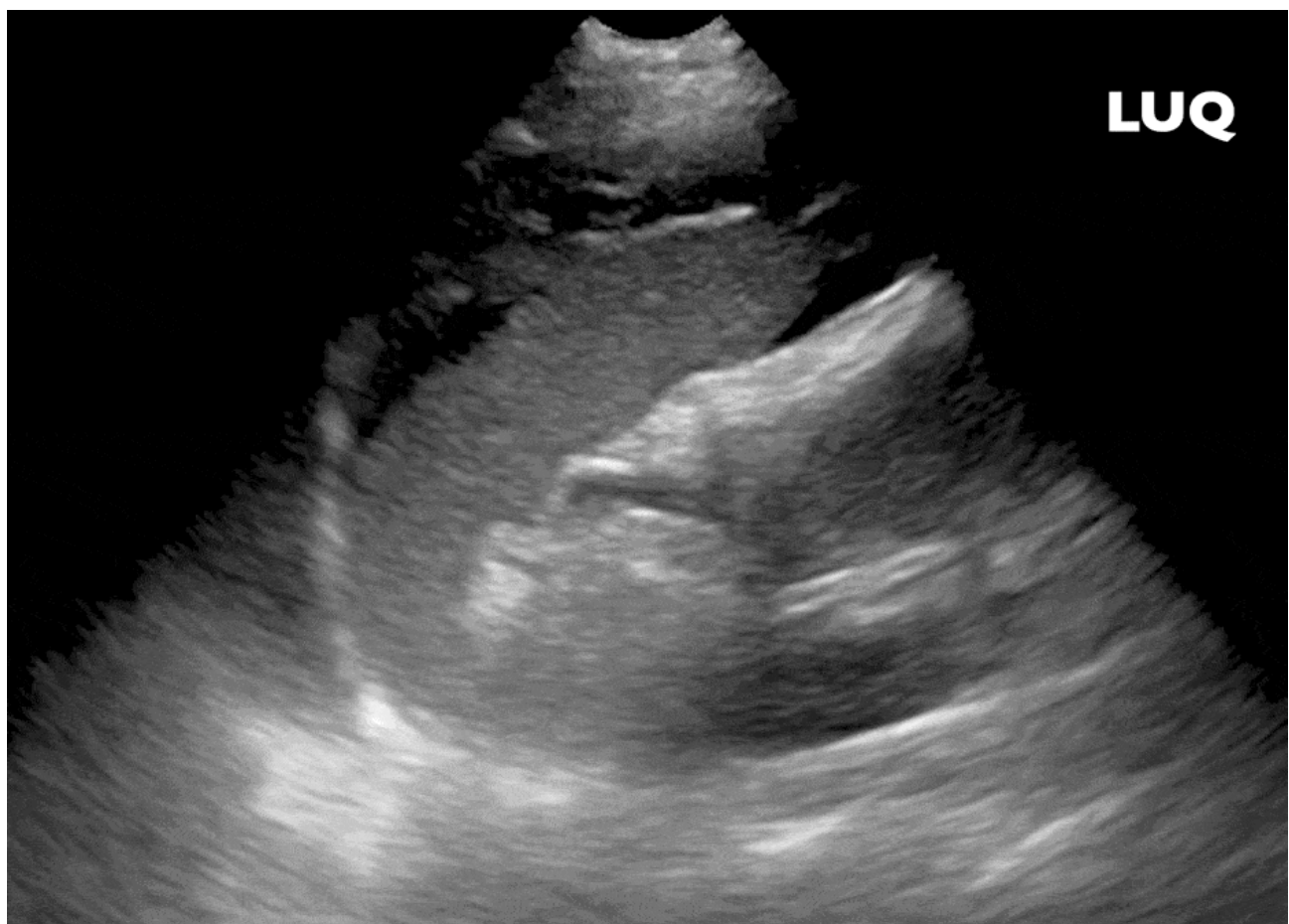
We will evaluate the LUQ in the eFAST for free fluid in the following places:

- **Perisplenic Space**
- **Spleen Tip**
- **Splenorenal Recess**

*POCUS 101 TIP: It is important to note that in the LUQ the most common area to find fluid is in the perisplenic space, NOT between the spleen and the left kidney. This is because there is a splenorenal ligament that attaches the spleen and the left kidney preventing a significant amount of fluid to accumulate there unless the ligament is ruptured.*



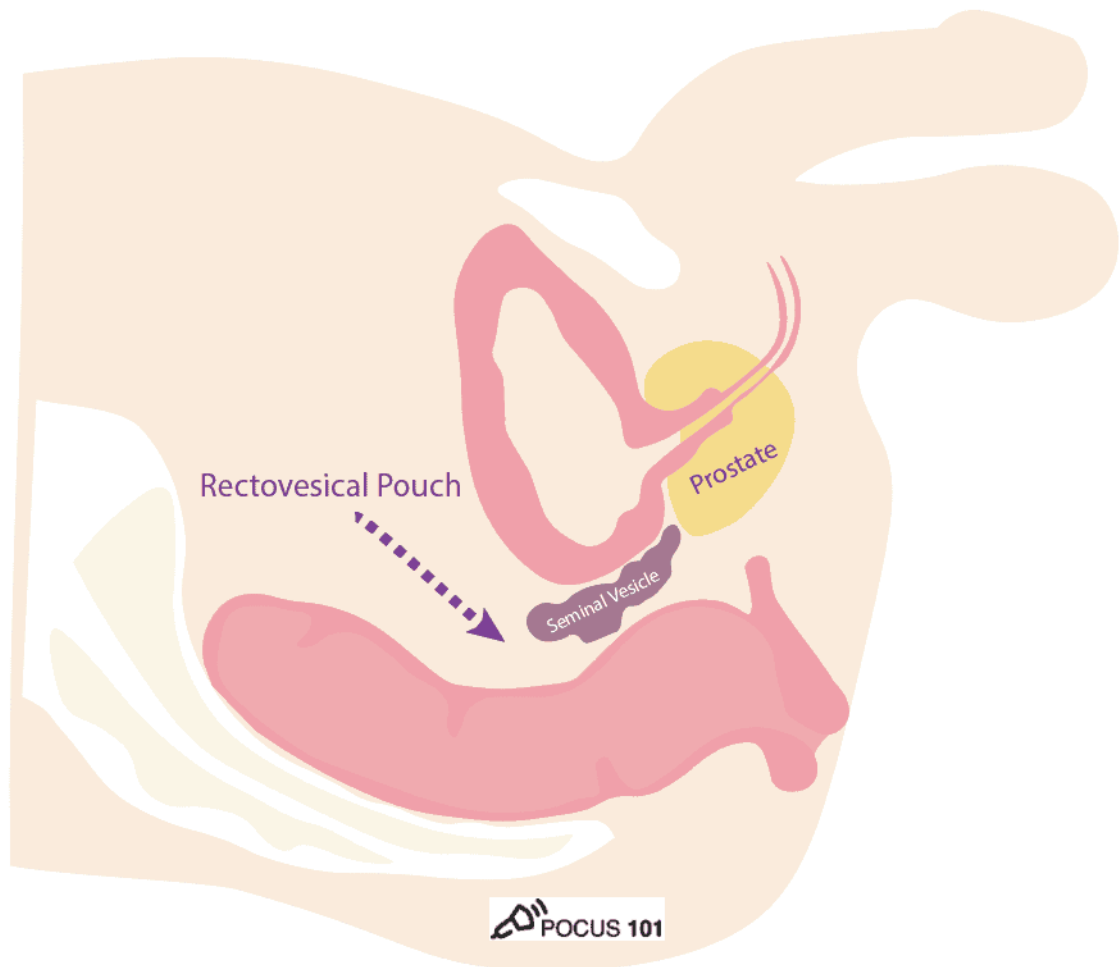
eFAST LUQ Perisplenic Space



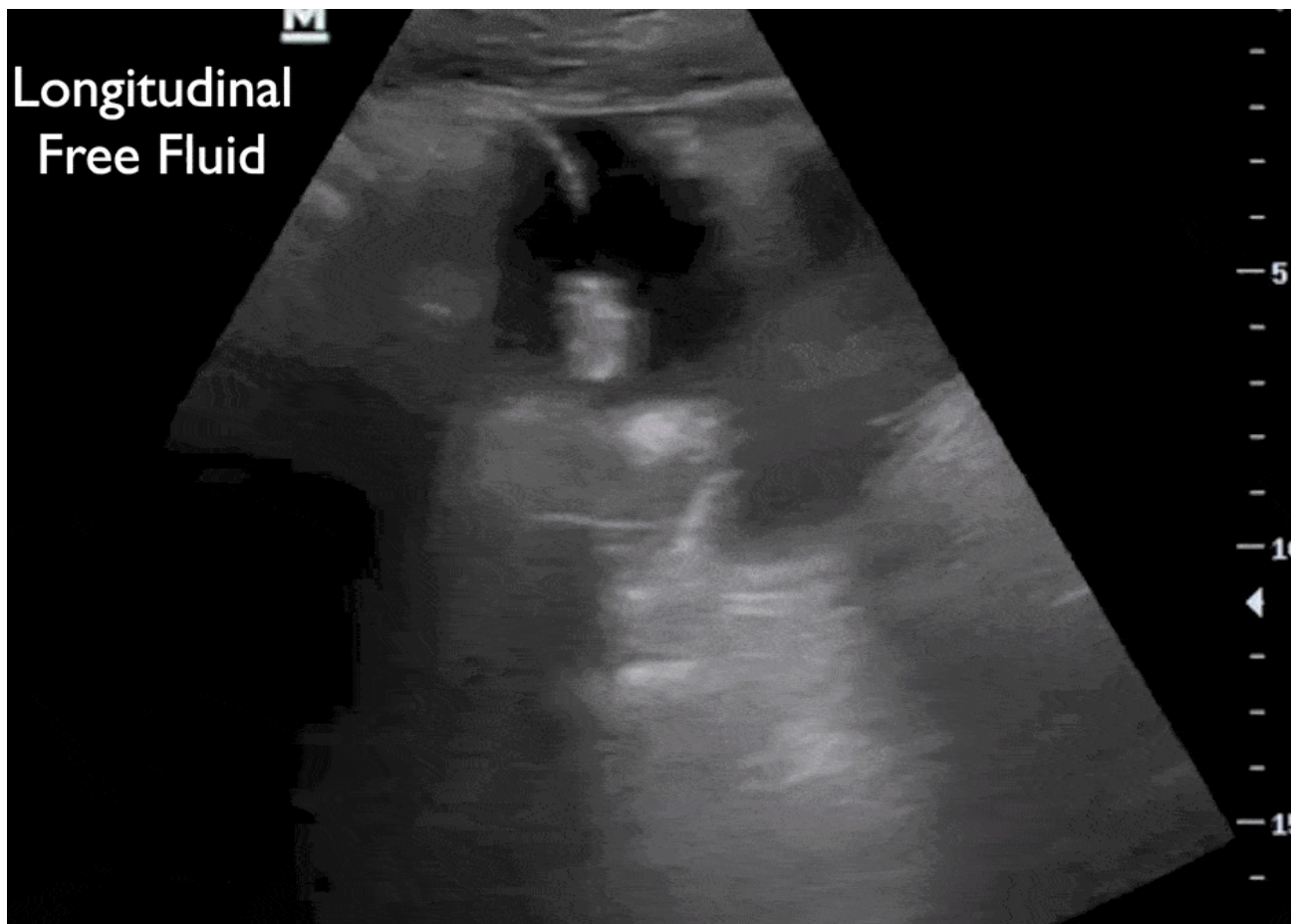
Free fluid in Perisplenic Space

## Male Pelvis – Hemoperitoneum

In the male pelvis, you can find free fluid in the rectovesical pouch/space.



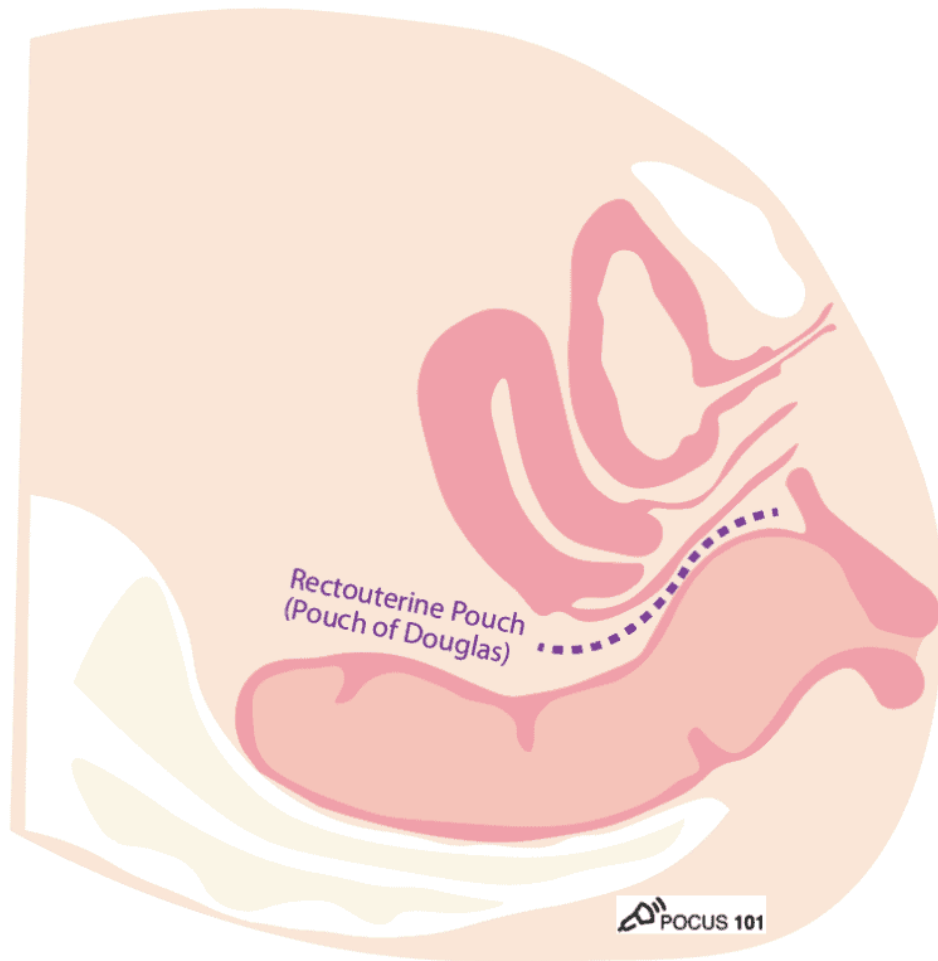
Male Pelvis Rectovesical Pouch



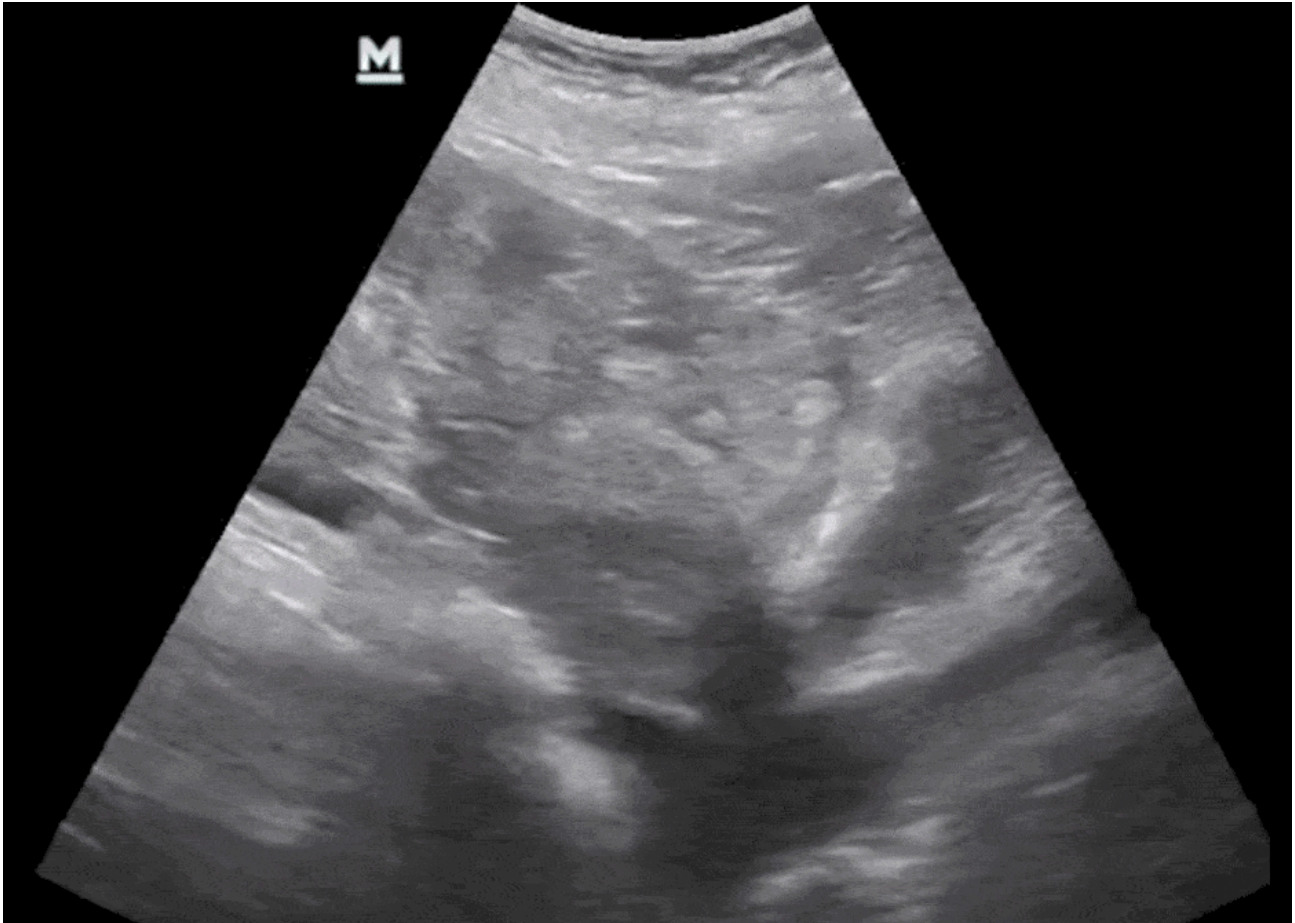
Male – Abdominal Free Fluid in Pelvis (Rectovesical Pouch)

#### **Female Pelvis – Hemoperitoneum**

In the female pelvis, you can find free fluid in the Pouch of Douglas (Rectouterine Pouch).



Female Pelvis Rectovesical Pouch



Female – Abdominal Free Fluid in Pelvis (Pouch of Douglas)

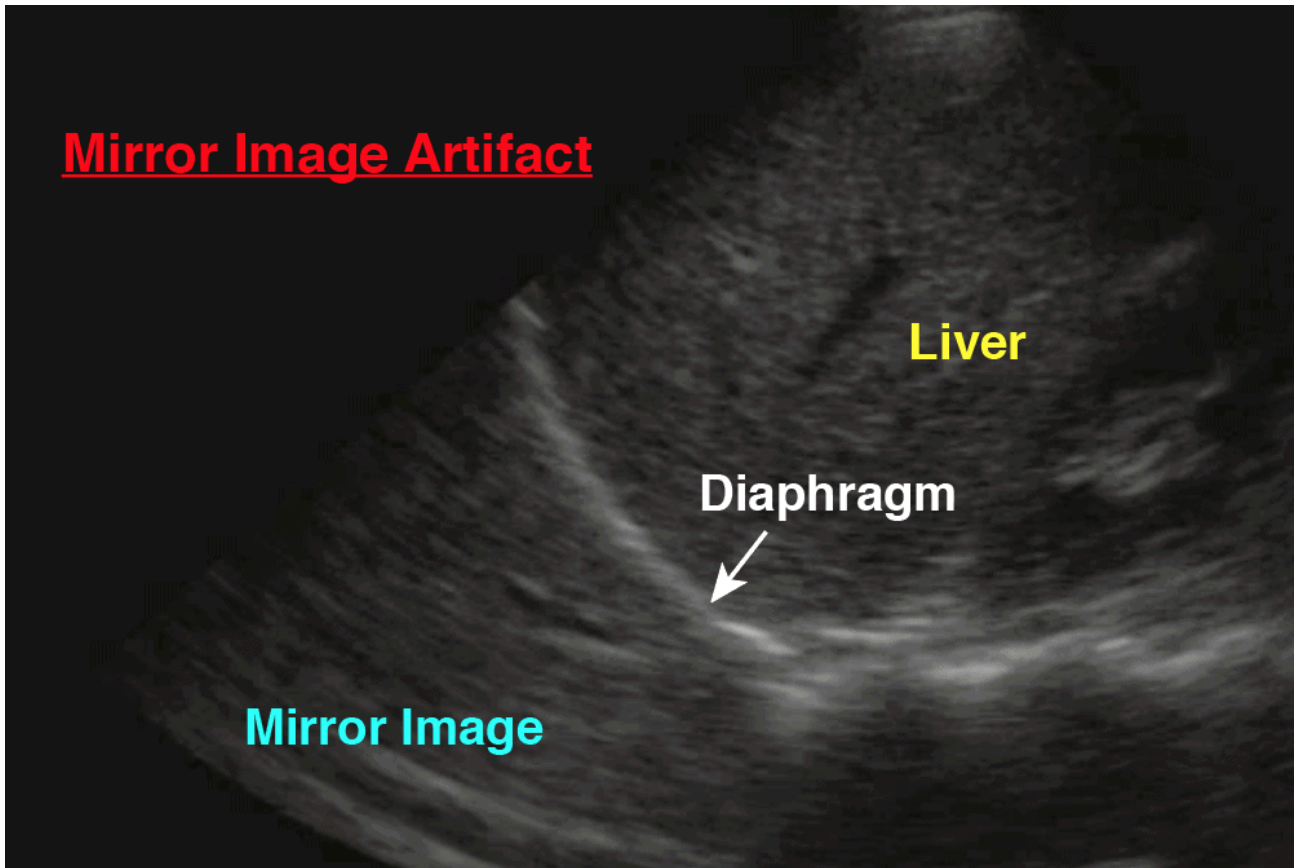
## Hemothorax

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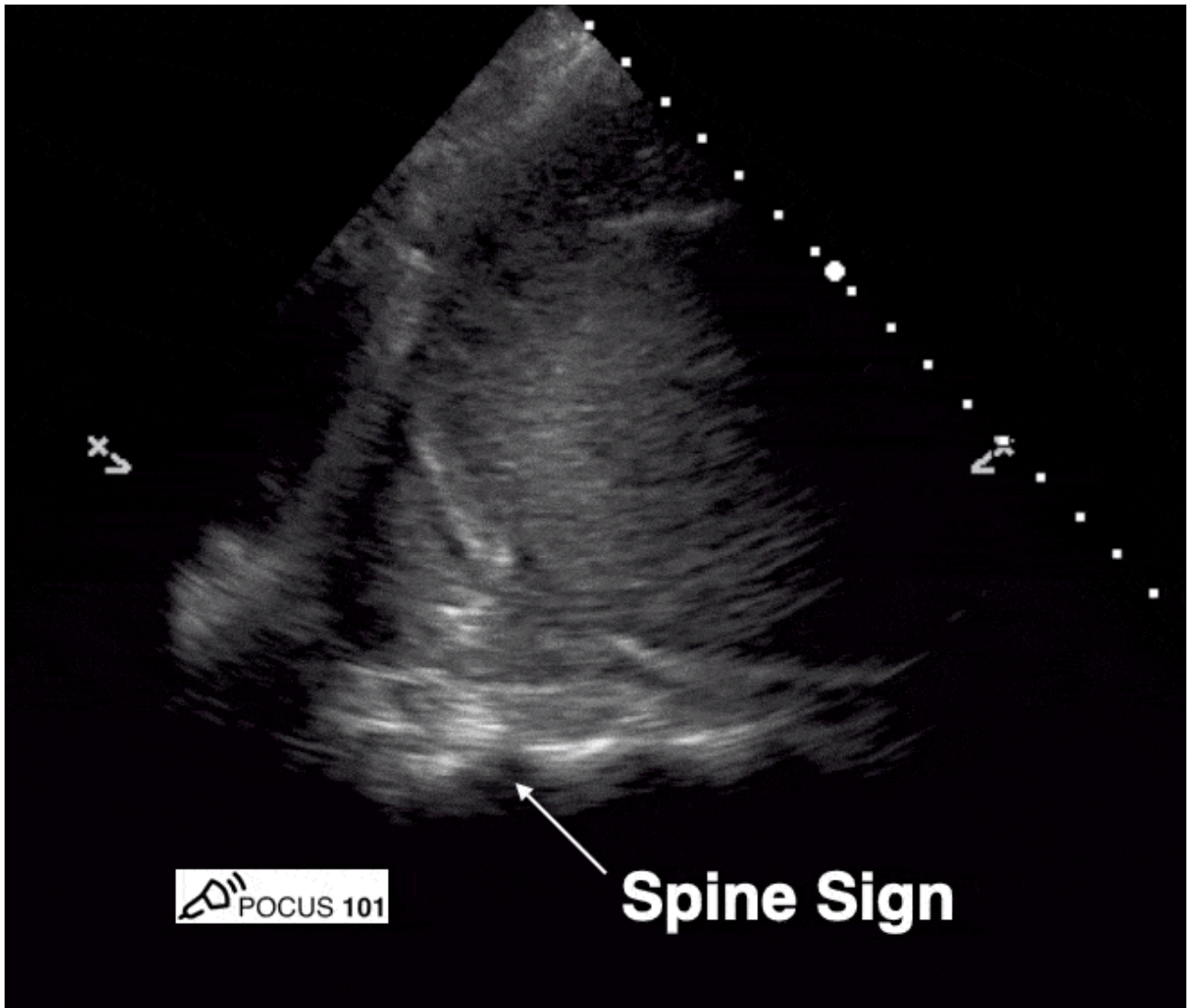
After evaluating the RUQ or LUQ, move the probe superiorly one or two rib spaces to evaluate the thorax for fluid accumulation.

A normal lung will have a Mirror Image Artifact and you will be unable to see the spine going above the diaphragm since all of the ultrasound waves will be reflected back by the aerated lung.

**Visualizing** the patient's spine **above the diaphragm** implies that there is **free fluid (e.g. blood) in the thorax** since ultrasound waves can easily pass through the free fluid in the chest cavity, allowing you to see the spine. This is referred to as a **Positive Spine Sign** ([click here](#) for a more in-depth explanation of the spine sign).



Absence of Spine above Diaphragm: **Normal Finding.**

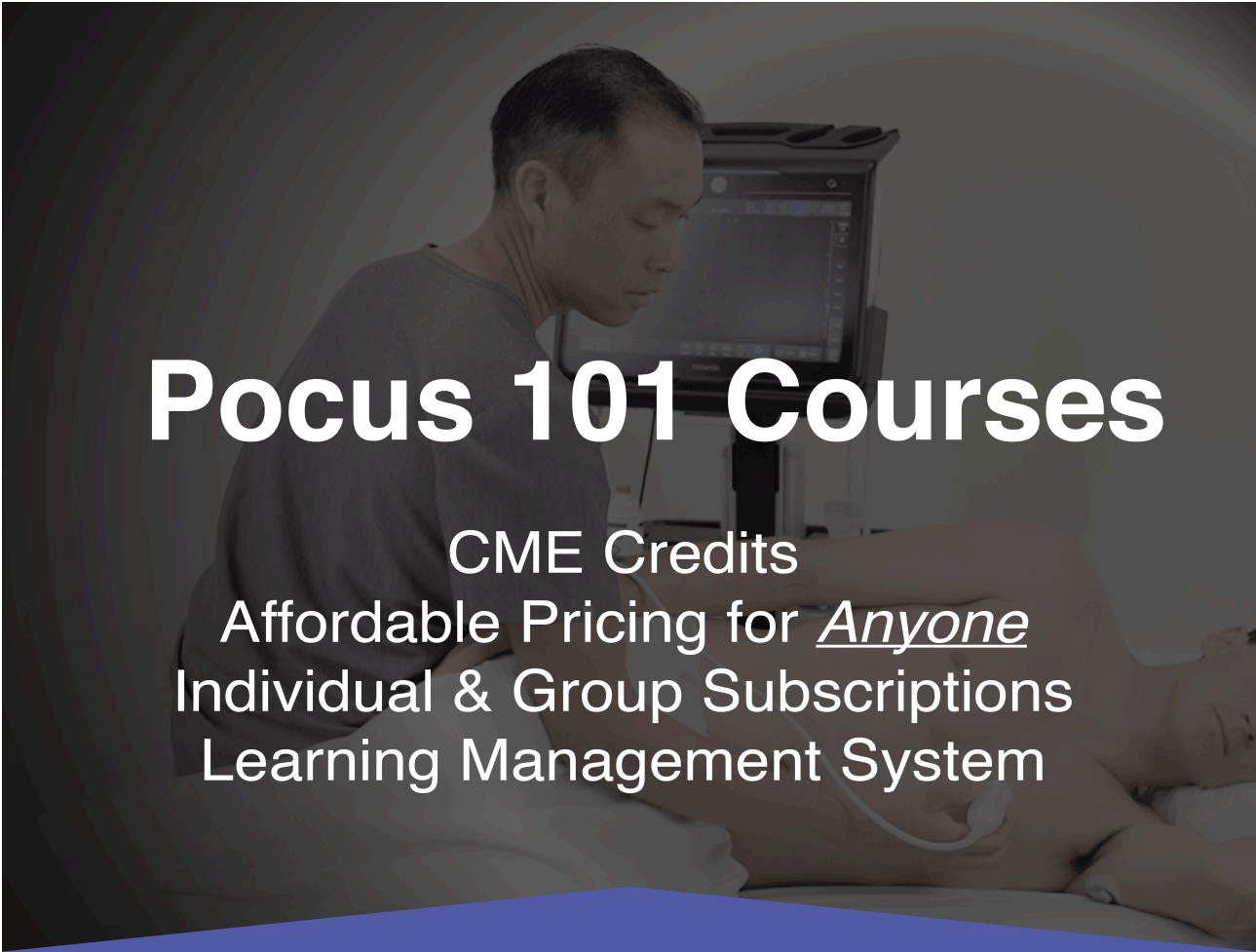


Presence of Spine above Diaphragm: **Pathologic Finding.**

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## Step 4: Aorta

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The next step is to evaluate for Abdominal Aortic Aneurysm (AAA) or Aortic Dissection as sources of hypotension in the RUSH Exam.

[Click Here for a step-by-step tutorial on how to perform Aorta Ultrasound!](#)

### Abdominal Aortic Aneurysm (AAA)

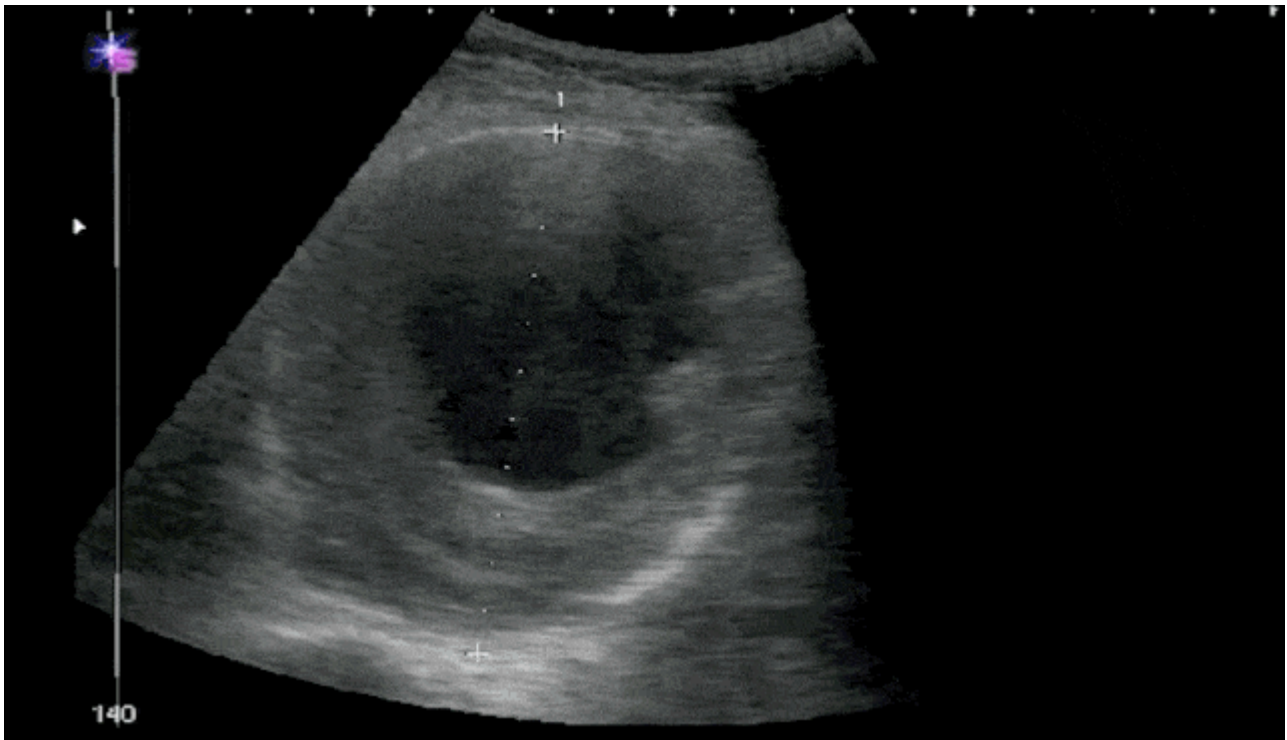
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A ruptured abdominal AAA can lead to severe hypotension, especially in elderly patients presenting with acute abdominal pain. A ruptured AAA has a very high mortality.

A normal aorta is usually **~2.0cm** in diameter. An abdominal aortic aneurysm is defined as: (Mokashi)

- **≥ 3cm diameter** for the **abdominal aorta** or a **> 50% increase** in the aortic diameter.

- **$\geq 1.5\text{cm}$  diameter** for the **iliac arteries**.
- Anytime a patient presents with a AAA of  $\geq 5\text{cm}$  and hypotension, assume a rupture until proven otherwise.
- Be sure to measure **Outer wall to Outer wall** for accurate aorta measurements.



Abdominal Aortic Aneurysm with Mural Thrombus

## Aortic Dissection

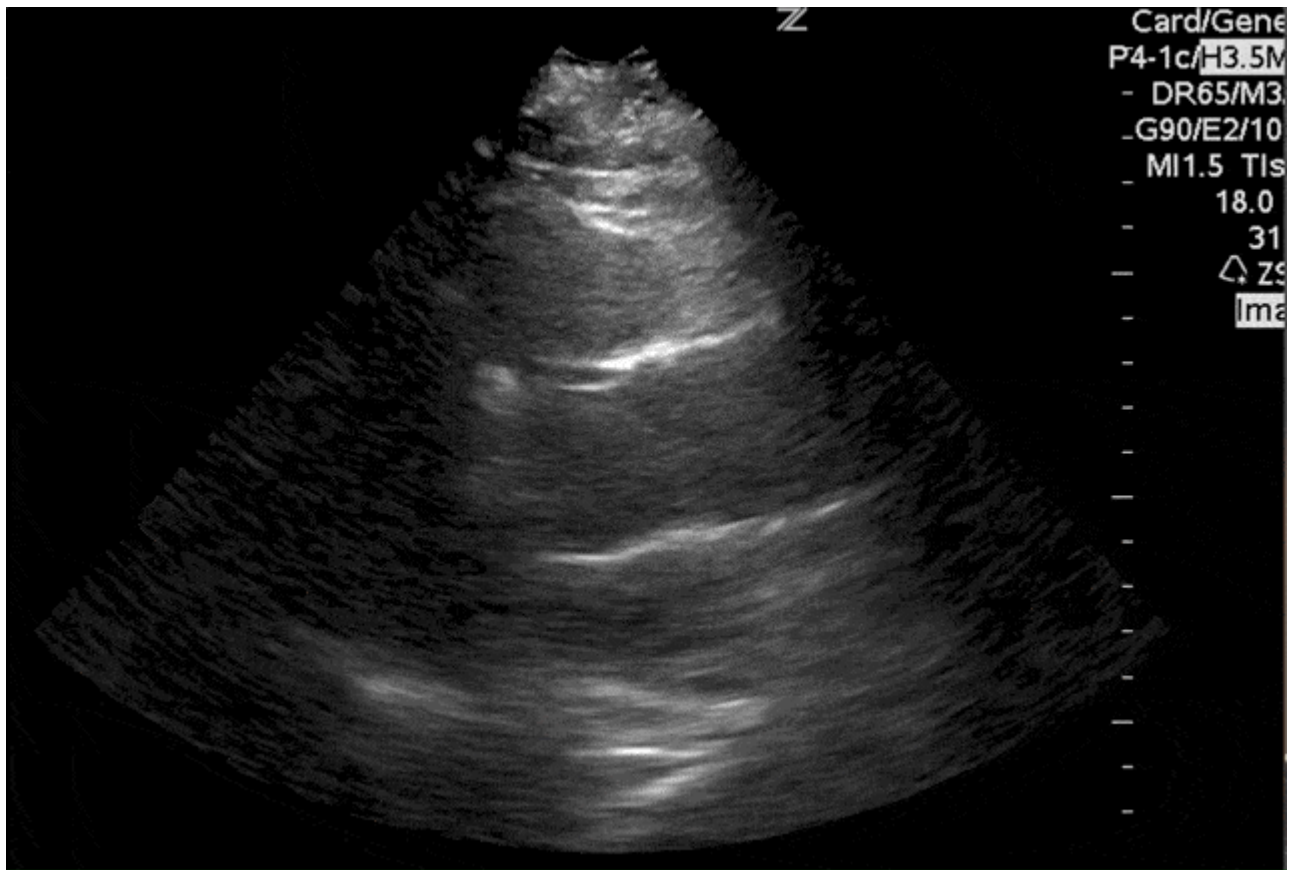
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Aortic dissections also have a very high mortality rate. The difficulty with diagnosing aortic dissections is that clinically, they can present with a wide array of symptoms including chest pain, abdominal pain, back pain, stroke-like symptoms, or even just generalized malaise.

Transthoracic Echocardiography (TTE) can be used to try to detect dissections and are specific but not very sensitive.

If there is a high clinical suspicion for aortic dissection and the TTE is negative, you may need to proceed with Transesophageal Echocardiography (TEE) or a CT angiogram (if patient is stable enough).

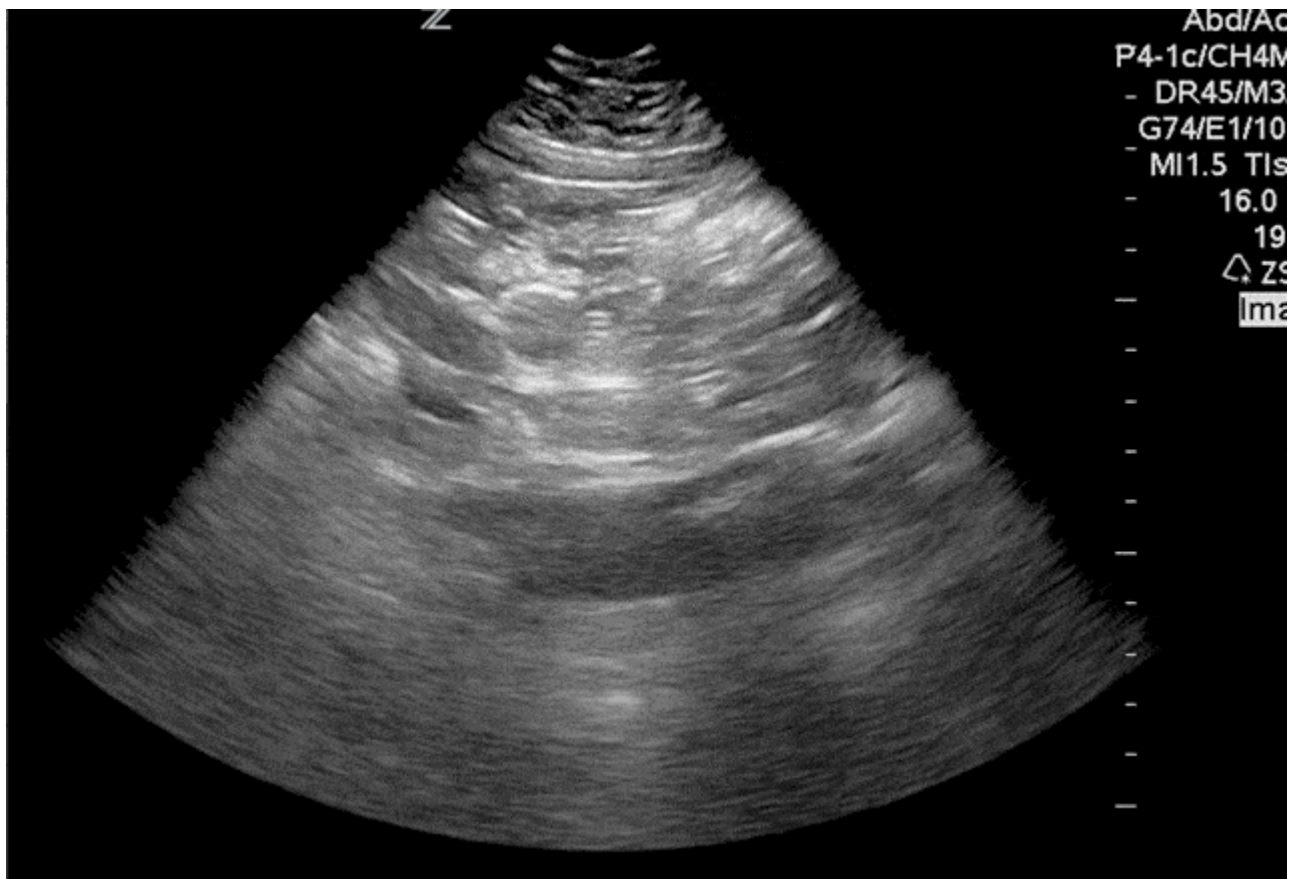
An aortic dissection may present as a free flap in the aortic lumen of either the descending abdominal aorta, ascending aorta, and/or the aortic arch. Aortic dissections in the ascending aorta can also cause aortic regurgitation and a diastolic murmur. Below are some examples of aortic dissections in multiple different views.



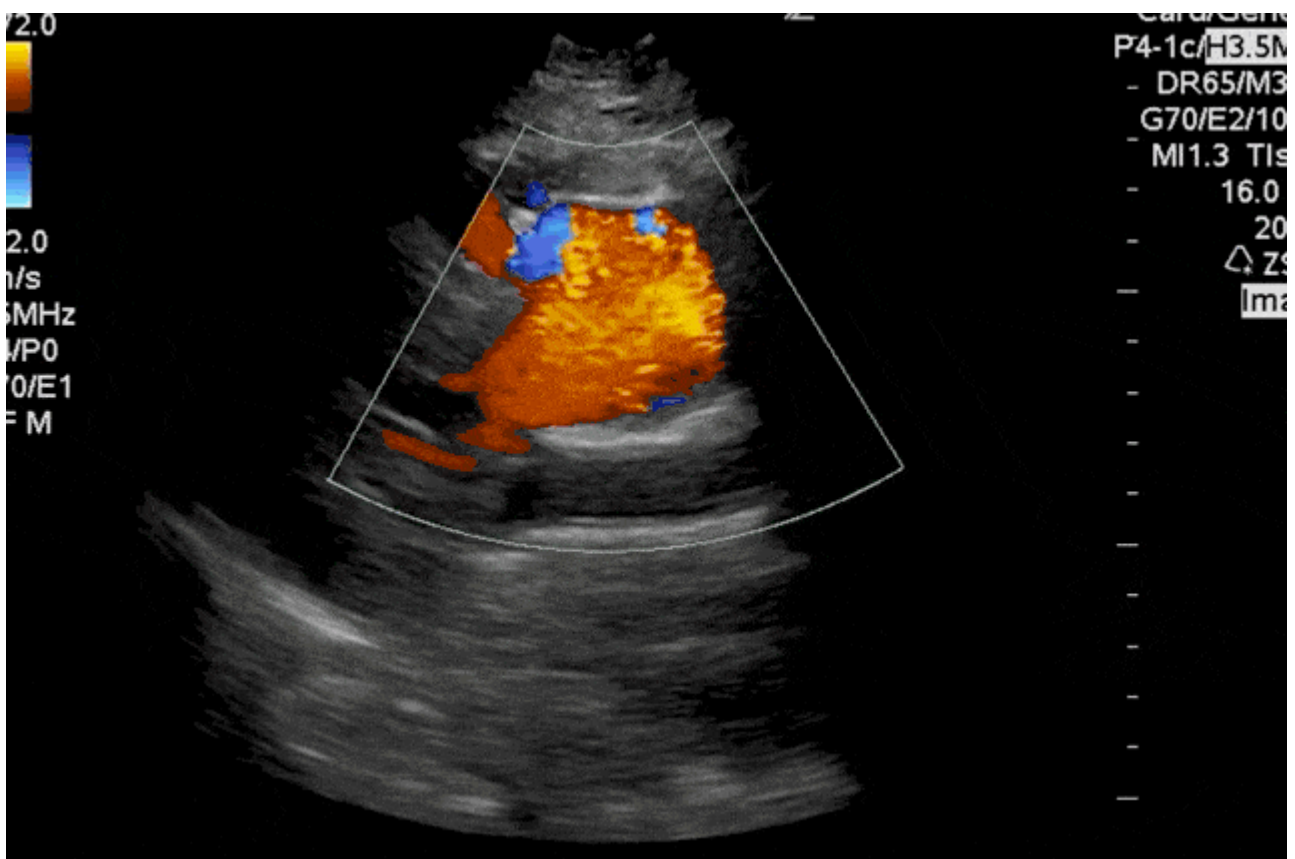
Aortic Dissection in the ascending aorta, seen as a longitudinal intimal flap just distal to the aortic valve in the PSLA View.



Aortic Dissection seen in the aortic arch using the Suprasternal View



Aortic Dissection with intimal flap seen in a long-axis view of the descending aorta.



Aortic regurgitation, seen as retrograde diastolic blood flow in the aortic arch.

## Step 5: Pulmonary

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For the last step of the RUSH exam, we look at the lungs or pulmonary system as a cause of hypotension/shock. Although we may have already identified pleural effusion or hemothorax on our RUQ/LUQ exam, this part of the lung exam will evaluate for a potential tension pneumothorax.

[Click Here for a step-by-step tutorial on how to perform Lung Ultrasound!](#)

### Pneumothorax

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**Here are three important steps to evaluating for pneumothorax:**

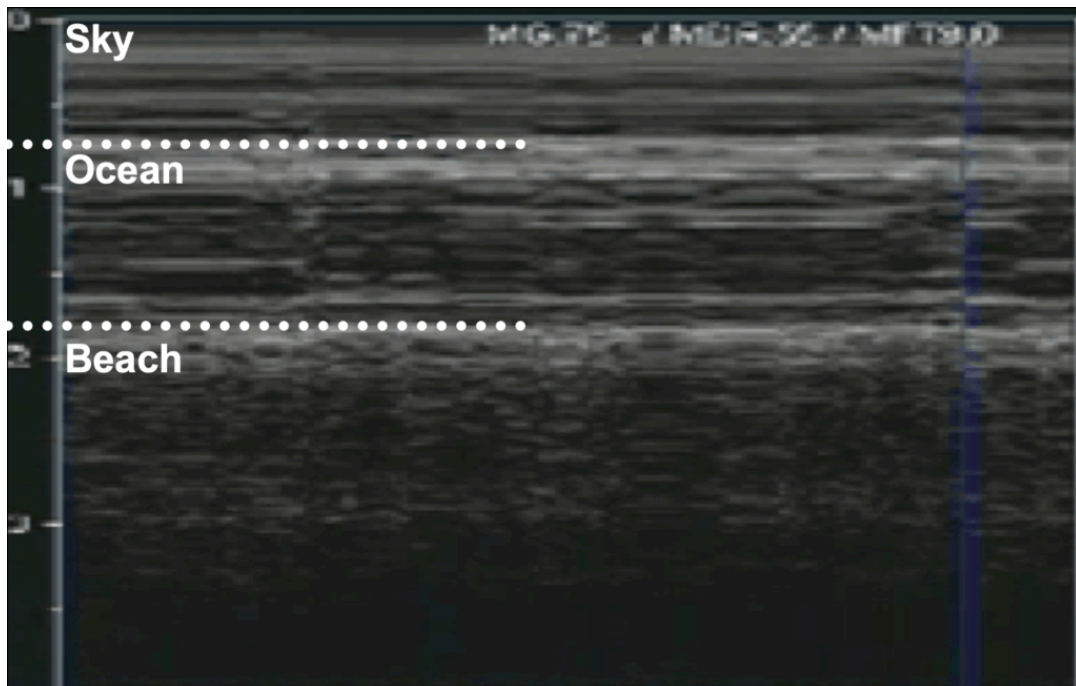
**First, if lung sliding is present, you can rule *out* pneumothorax with 100% accuracy at that ultrasound point** (Husain LF).

Remember that presence of lung sliding only rules out pneumothorax at that specific point you are scanning. Make sure to maximize your sensitivity by scanning multiple points on the chest.

You can look for lung sliding with B-mode or M-mode:



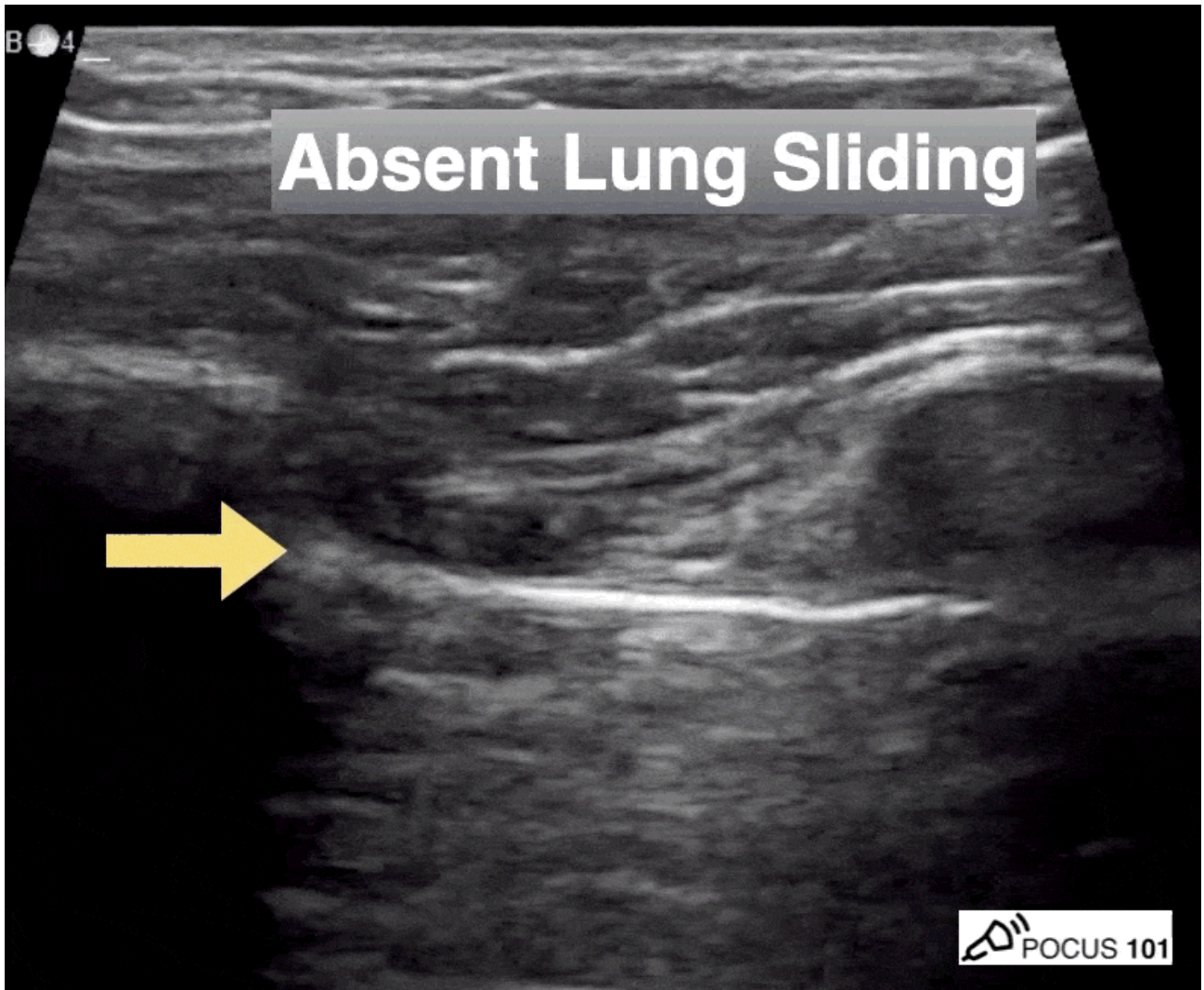
Normal Lung Sliding (B-mode)



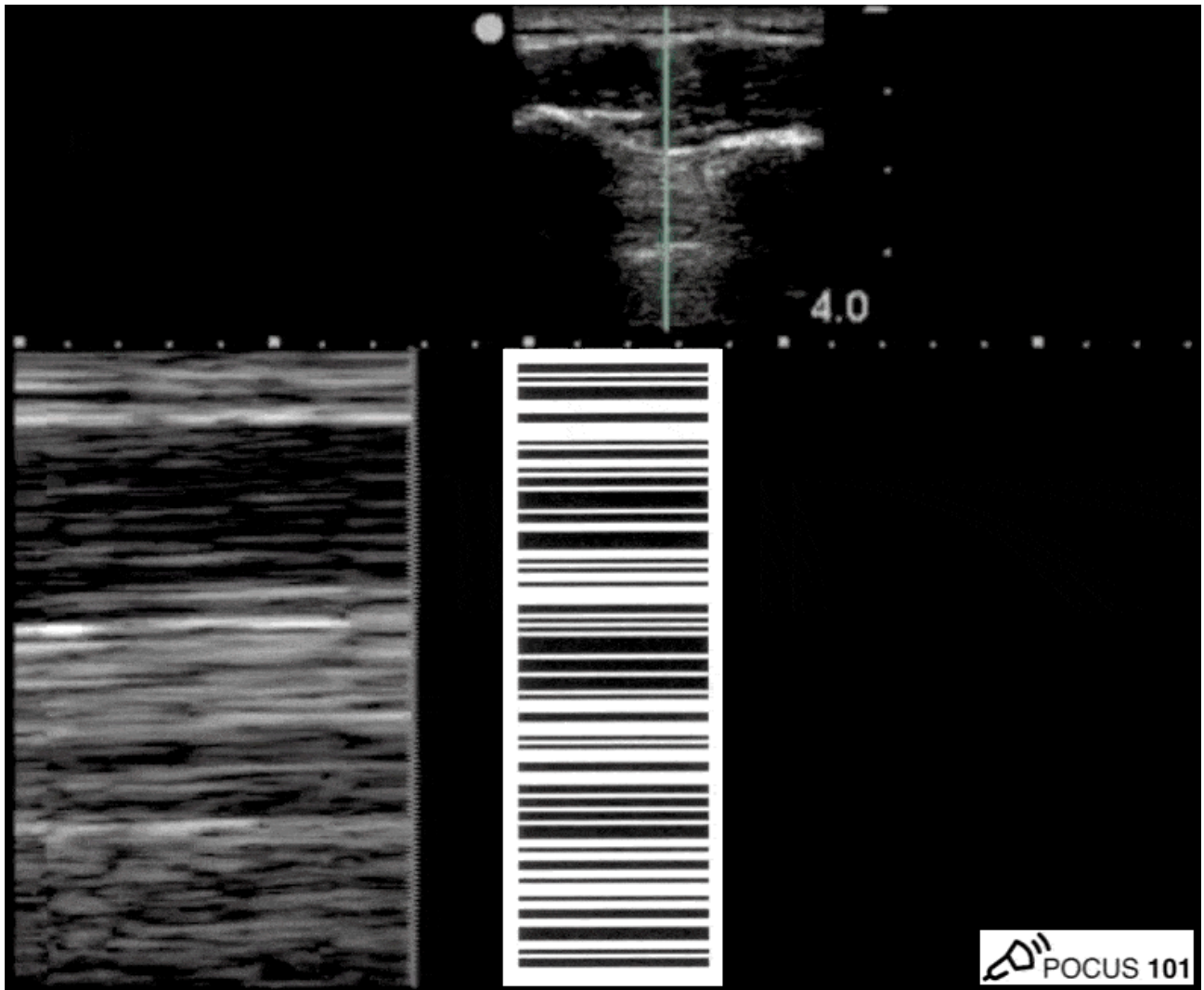
Normal Lung Sliding with Seashore sign (M-mode)

**Second, if lung sliding is **ABSENT**, you should *not automatically assume* pneumothorax.**

Recall other causes of reduced/absent lung sliding: severe consolidation, chemical pleurodesis, acute infectious or inflammatory states, fibrotic lung diseases, acute respiratory distress syndrome, or mainstem intubation.



Absence of Lung Sliding (B-mode)



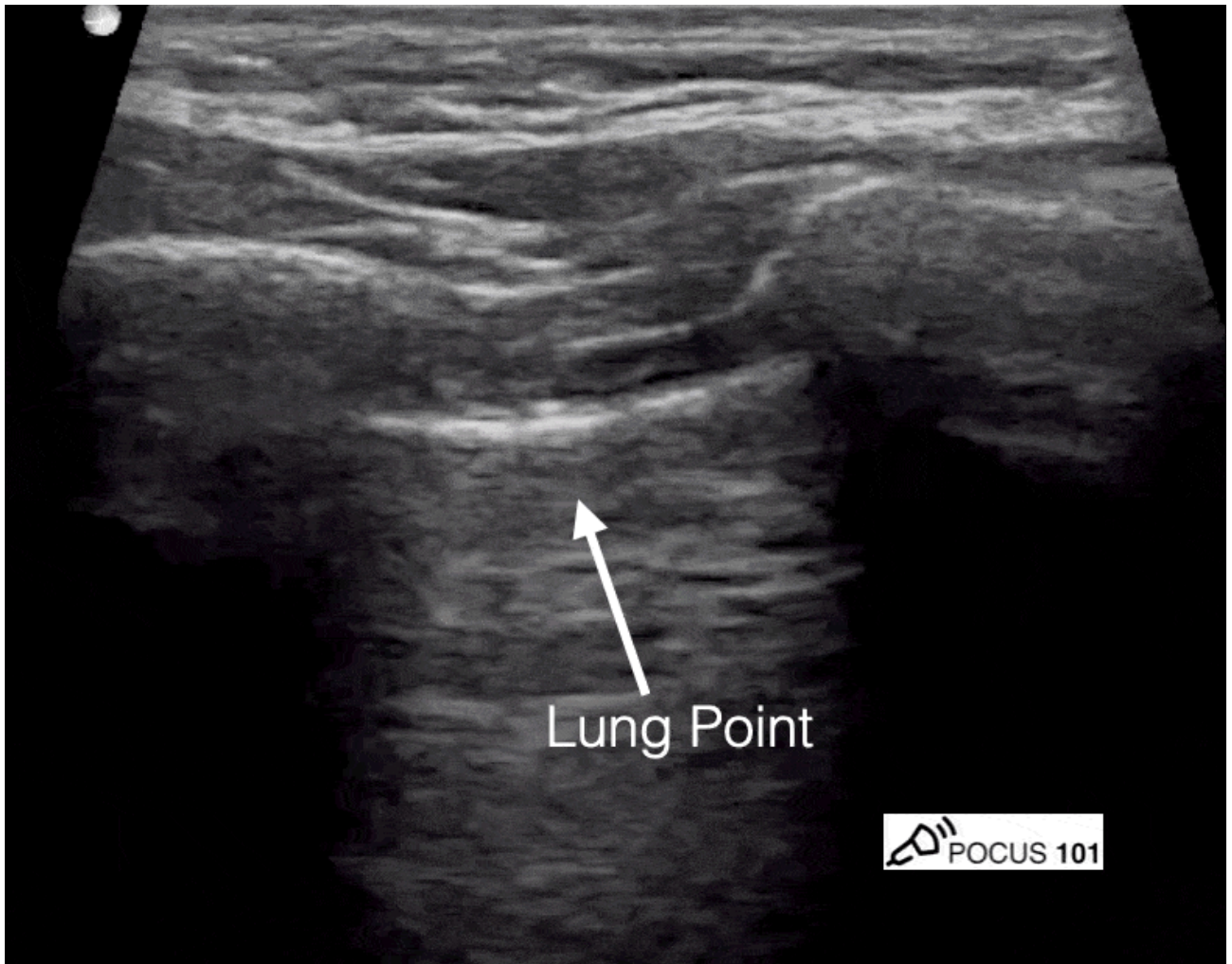
Absence of Lung Sliding – Barcode Sign (M-Mode)

**Third, if a lung point is present, you can rule *in* pneumothorax with 100% accuracy** (Chan S).

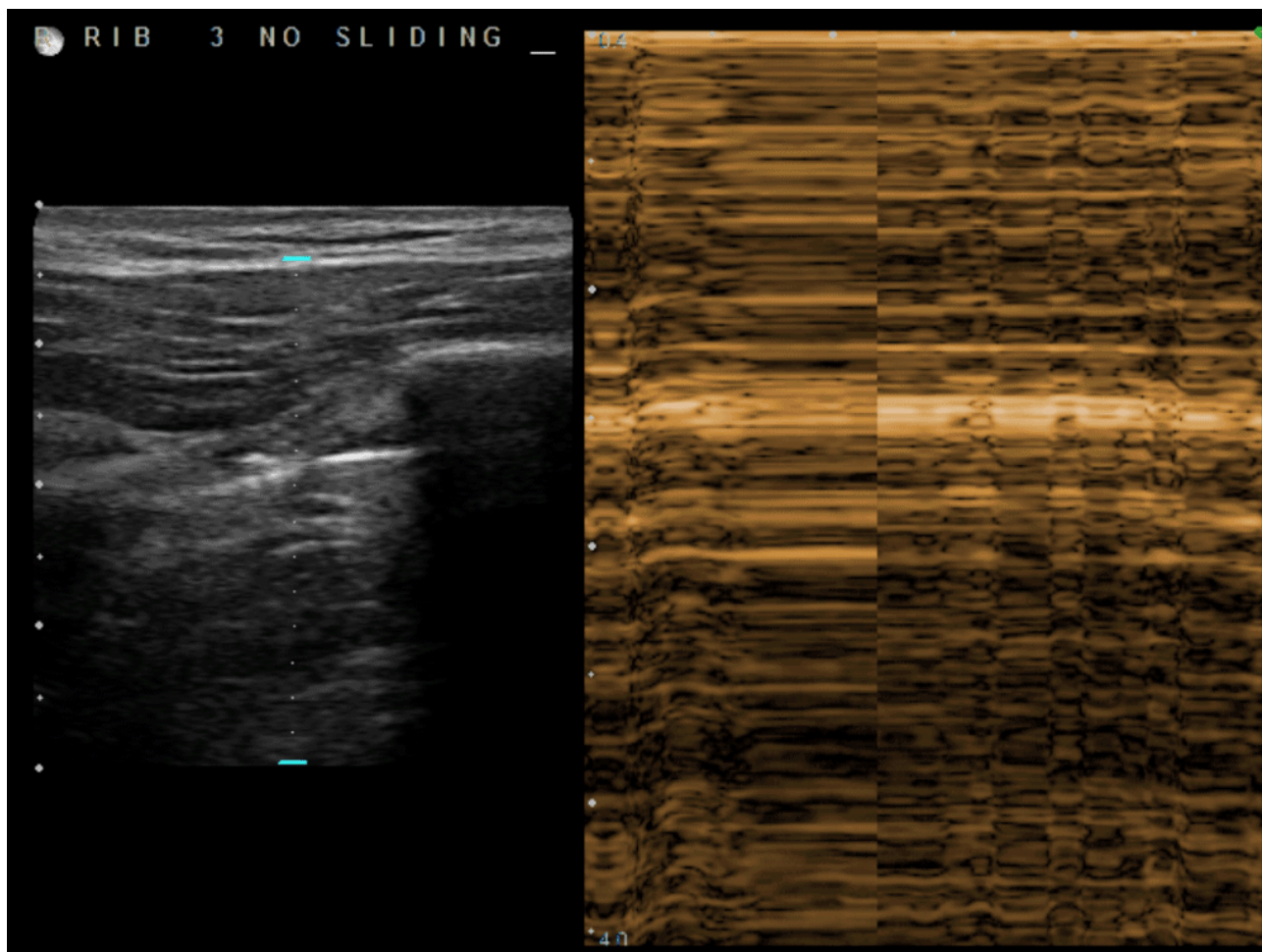
To confirm the presence of a pneumothorax, you should look for the “**Lung Point Sign.**”

The **lung point** is when you can see the transition between normal lung sliding and the absence of lung sliding. This is the transition point between the collapsed lung and normal lung. If you see this you can definitively *rule in* a pneumothorax. The Lung point sign also helps you quantify how large a pneumothorax is.

If you think you may have found a lung point but are not sure, use M-Mode and place your cursor at the intersection where you think lung sliding starts and stops. If you see a normal seashore sign that turns into an abnormal barcode sign, then you have located the lung point with M-Mode.



Lung Point Sign (B-mode)



Lung Point Sign (M-mode)

And that's it for the RUSH Exam Ultrasound Protocol! Let us know what you think in the comments section.

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